HEMORA - IDENTIFICATION OF HEMATOLOGICAL DISEASES USING DEEP LEARNING AND MICROSCOPIC BLOOD SMEAR IMAGE PROCESSING

25-26J-514

Project Proposal Report

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Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Acute Lymphoblastic Leukemia (ALL) diagnosis in Sri Lanka faces significant challenges due to limited laboratory facilities, 3-7 day delays in test results, and shortage of specialist hematologists, with traditional manual microscopic examination being time-intensive and error-prone. This research focuses on developing the ALL detection component of HEMORA, a comprehensive hematological disease identification system, by creating a hybrid deep learning framework that integrates CNN-YOLO architecture for peripheral blood smear image analysis with machine learning algorithms for Complete Blood Count (CBC) parameter analysis to accurately identify and classify lymphoblasts in Sri Lankan patients. The methodology employs decision-level fusion combining morphological features (large nuclei, high nuclear-to-cytoplasmic ratio) detected through image processing with quantitative CBC parameters (WBC count, lymphoblast percentage, hemoglobin levels) to enhance diagnostic accuracy for ALL detection. Anticipated test results include achieving 92-95% accuracy in lymphoblast detection with sensitivity and specificity exceeding 90%, reducing diagnostic time from 3-7 days to under a minimum time, and demonstrating superior performance compared to single-modal approaches through validation with real-world clinical data from Sri Lankan hospitals. This research contributes the ALL detection module to the broader HEMORA system, concluding that hybrid AI approaches can effectively address diagnostic limitations in resource-constrained healthcare environments, with recommendations for clinical integration and deployment as a web-based diagnostic support tool for healthcare providers.

Keywords: Acute Lymphoblastic Leukemia, Deep Learning, Medical Image Processing, Hematological Diagnosis, Healthcare AI

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ALL	Acute Lymphoblastic Leukemia
AWS	Amazon Web Services
CBC	Complete Blood Count
CNN	Convolutional Neural Network
GPU	Graphics Processing Unit
HIS	Hospital Information Systems
IDA	Iron Deficiency Anemia
IT	Information Technology
ML	Machine Learning
NLP	Natural Language Processing
PBS	Peripheral Blood Smear
RAM	Random Access Memory
RBC	Red Blood Cell
SCD	Sickle Cell Disease
WBC	White Blood Cell
YOLO	You Only Look Once

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1. INTRODUCTION

1.1 Background

There is an increase in hematological illnesses in Sri Lanka due to a lack of diagnostic facilities, laboratory delays, and a shortage of specialists, all of which prevent early diagnosis based on accuracy and timing. Acute Lymphoblastic Leukemia (ALL), Thalassemia, Iron Deficiency Anemia (IDA) and Sickle Cell Disease (SCD) have serious consequences for public health, causing severe complications and increased mortality if not discovered early [1]. Blood test results at Sri Lanka's government hospitals take 3 to 7 days, and rural patients must travel long distances for specialist hematological testing [2,3]. As a result, automated, efficient, and accessible diagnostic solutions adapted to the Sri Lankan healthcare system are a top focus.

This problem is resolved by developing an AI-powered system capable of detecting and recognizing the four major hematological diseases—Acute Lymphoblastic Leukemia (ALL), Thalassemia, Iron Deficiency Anemia (IDA), and Sickle Cell Disease (SCD)—from blood smears and CBC results. The proposed solution, which employs advanced image processing and deep learning algorithms, is expected to improve the availability of modern healthcare systems as well as Sri Lanka's healthcare infrastructure by reducing diagnostic time, advanced healthcare costs, human error, and the country's advanced healthcare system and infrastructure.

As a subset of this larger research area, my role is to identify and detect Acute Lymphoblastic Leukemia (ALL). ALL is a cancerous blood disorder caused by the abnormal proliferation of immature white blood cells (lymphoblasts), which interfere with normal blood production. Children's ALL incidence in Sri Lanka continues to rise, with reduced survival rates caused by delayed diagnosis [4]. To combat this, my contribution is the creation of a deep learning model for identifying lymphoblasts in peripheral blood smear images by means of preprocessing (resizing, contrast stretching, and segmentation). Deep classification approaches like CNNs, YOLO, or Vision Transformers are to be tried out while a hybrid decision-making framework is integrated with CBC parameters like white blood cell count, hemoglobin level, and

lymphoblast percentage. This combined methodology should improve diagnostic accuracy, lower false positives/defaults, and give medical practitioners a web-app for rapid and precise detection of ALL in Sri Lankan hospitals [4,5].

1.2 Literature Survey

The conventional diagnosis of Acute Lymphoblastic Leukemia (ALL) is based on manual microscopic inspection of peripheral blood smears, which is difficult, time-consuming, and heavily reliant on a doctor's subjective assessment [1]. This dependency can cause severe diagnostic delays, especially in low-resource settings, negatively impacting patient outcomes [2,3].

To address these limitations, researchers have increasingly explored **automated image analysis using deep learning**. Convolutional Neural Networks (CNNs) have been widely applied for classifying lymphoblasts from blood smear images. Rehman *et al.* [4] developed a CNN-based framework that achieved high accuracy in identifying ALL, demonstrating the model's ability to capture intricate morphological features without manual feature engineering. Similarly, Talo [5] applied CNNs for automated blood cell classification, highlighting the efficiency and reliability of deep learning in hematological diagnostics.

Beyond classification, **object detection techniques** have been employed to not only identify but also localize lymphoblasts within blood smear images, mimicking a pathologist's workflow. Tlelo-Cuautle *et al.* [7] used a YOLO-based system to detect different types of white blood cells and ALL cells, providing precise localization that can enhance diagnostic confidence. A systematic review by Pansini *et al.* [6] summarizes these deep learning and computer vision approaches, noting that while image-based models achieve high accuracy, they often overlook other diagnostic data that could improve robustness.

An emerging approach in the field is **hybrid or multi-modal systems** that combine blood smear image analysis with quantitative hematological data, such as Complete Blood Count (CBC) parameters. The rationale is that ALL typically presents with

distinct blood profiles, including elevated white blood cell counts, anemia, and thrombocytopenia. Such hybrid methods integrate visual morphology with numerical blood data, potentially improving diagnostic accuracy and reducing false positives [9].

In the Sri Lankan context, research on automated ALL detection remains limited. While studies have applied machine learning to CBC data for predicting other hematological conditions, such as anemia [9], there is no published work on integrated or hybrid systems for ALL detection tailored to local patient populations [2]. Furthermore, most international models are trained on datasets that may not reflect the genetic and environmental diversity of Sri Lankan patients, which can limit their applicability and accuracy within the local healthcare system [3,6].

This review of prior work highlights a clear gap: while global research demonstrates the promise of CNNs, YOLO, and hybrid systems, there is a lack of **context-specific, multi-modal ALL detection systems** for Sri Lanka. This motivates the present study, which proposes a novel hybrid deep learning framework combining CNN- or YOLO-based blood smear image analysis with machine learning models interpreting CBC parameters, aiming to deliver faster, more accurate, and locally relevant diagnostics for ALL patients in Sri Lanka.

1.3 Research Gap

Despite significant growth in AI-assisted hematological diagnoses, there are still a number of crucial gaps, particularly when considering Sri Lanka.

- **Absence of Multi-modal ALL Detection Systems**

Current research approaches ALL detection through isolated methodologies either blood smear image analysis [4,5,7,8] or CBC parameter analysis [9] but lacks integration of both modalities. While Rehman *et al*. [4] achieved high accuracy using CNNs for lymphoblast classification and Herath *et al*. [9] successfully applied machine learning to CBC data in Sri Lanka, no existing literature demonstrates a unified framework that combines morphological features with quantitative blood parameters for comprehensive ALL diagnosis.

- **Limited Population-Specific Validation for Sri Lankan Patients**

Most research [4,5,6,7,8] use datasets from Western or non-South Asian populations, resulting in a significant representational gap for Sri Lankan patients. Given the well-documented genetic, environmental, and lifestyle variables that determine hematological profiles, existing AI models may not be applicable to South Asian contexts. No prior study has been undertaken to design or validate ALL detection systems using Sri Lankan patient data, which is required for reliable and clinically relevant diagnostic tools.

- **Lack of Hybrid AI Architectures for Enhanced Accuracy**

The current literature primarily uses single-model techniques for object identification, such as CNNs [4,5,8] or YOLO [7]. While each achieves acceptable performance, no study has looked into hybrid AI architectures that combine CNNs, object detection models (e.g., YOLO), and standard machine learning for CBC analysis into a single decision-support system. Such hybridization could reduce false positives and negatives while achieving diagnostic accuracy comparable to experienced hematologists.

- **Insufficient Consideration of Healthcare Infrastructure Integration**

Although Pansini *et al*. [6] highlight strong laboratory accuracy, their review overlooks practical deployment challenges in resource-limited healthcare settings. In Sri Lanka, where blood test results can be delayed by 3–7 days [2] and rural patients often travel long distances for consultation [3], existing solutions fail to address infrastructural and accessibility constraints. No study provides strategies for low-resource deployment, offline operation, or seamless integration into existing clinical workflows in developing countries.

- **Absence of Real-Time Clinical Decision Support Tools**

The majority of studies emphasize accuracy metrics but lack translation into real-time diagnostic interfaces. None has introduced practical tools capable of immediate ALL detection with confidence scoring and lymphoblast localization. Such tools are crucial

for empowering non-specialist healthcare workers to make informed clinical decisions quickly and reliably.

To clearly highlight the shortcomings of existing studies and the novelty of the proposed research, Table 1.3.1 presents a feature-wise comparison of prior works. It demonstrates how current literature addresses certain aspects of ALL detection while leaving critical elements unaddressed, all of which are incorporated into the proposed framework.

Table 1.1: Critical Missing Elements in Existing Research

Missing Elements	**[4]**	**[5]**	**[6]**	**[7]**	**[8]**	**[9]**	**Proposed**
Multi-modal Data Integration	No	No	No	No	No	No	**Yes**
Sri Lankan Population Focus	No	No	No	No	No	Yes	**Yes**
Hybrid AI Architecture	No	No	No	No	No	No	**Yes**
Real-time Web Interface	No	No	No	No	No	No	**Yes**
Clinical Decision Support	No	No	No	No	No	No	**Yes**
Object Localization + Classification	No	No	Yes	Yes	No	No	**Yes**
False Positive Reduction via CBC	No	No	No	No	No	No	**Yes**
Resource-Constrained Deployment	No	No	No	No	No	No	**Yes**

1.4 Research Problem

According to the gaps observed, the research problem can be defined as follows:

Even with significant advancements in AI-assisted diagnostic tools, the only ways to identify Acute Lymphoblastic Leukemia (ALL) at this time are through CBC parameter analysis [9] or peripheral blood smear image analysis [4,5,7,8]. Additionally, the majority of models were developed and verified using datasets from the West [4,5,6,7,8], which limited their applicability to the people of Sri Lanka. These

drawbacks, especially in healthcare settings with limited resources, impair practical clinical application, delay early diagnosis, raise false positives and negatives, and reduce diagnostic accuracy [2,3,6].

Integration Challenge:

How can CBC numerical data and peripheral blood smear images be integrated into a single AI framework to enhance diagnostic accuracy? Current studies rely solely on image-based CNN models [4,5,7,8] or machine learning on CBC indices [9], resulting in unrealized potential when these modalities are used in isolation.

Sri Lankan Population-Specific Model Development:

How can a hybrid AI system be trained and validated on Sri Lankan patient data to capture unique population-specific hematological variations? Models developed on international datasets [4,5,6,7,8] may not accurately reflect the genetic, environmental, and hematological characteristics of Sri Lankan patients [2,3,9].

Differential Diagnosis Capability:

How can the proposed hybrid framework effectively differentiate Acute Lymphoblastic Leukemia (ALL) from other leukemias as well as from Thalassemia (Minor), Iron Deficiency Anemia (IDA), and Sickle Cell Disease (SCD), thereby reducing misdiagnosis? Accurate discrimination is vital for treatment planning, particularly in separating ALL from hematologically overlapping conditions [4,5,6,9].

Web-Based Real-Time System:

How can a lightweight, web-based, real-time diagnostic system be designed for deployment in Sri Lankan hospitals? Such a solution must function with limited computational resources, provide immediate results, and reduce dependency on specialized hematologists, thereby increasing accessibility in rural and under-resourced regions [2,3,6].

Clinical Decision Support and Explainability:

How can the diagnostic framework provide localized lymphoblast detection, confidence scoring, and explanatory outputs to assist clinicians in validating AI

predictions? Current models focus only on accuracy metrics without offering clinically interpretable decision support [4,5,7,8].

To solve these challenges, a Sri Lanka-specific hybrid AI system that integrates CBC numerical data with blood smear image analysis is required, enabling accurate detection of Acute Lymphoblastic Leukemia, reliable differentiation from other leukemias as well as Thalassemia (Minor), Iron Deficiency Anemia (IDA), and Sickle Cell Disease (SCD), and practical deployment in resource-limited clinical environments [2,3,4,5,6,7,8,9]. This research will therefore focus on developing and implementing such a system, bridging the gap between current diagnostic limitations and the potential of AI-assisted ALL detection.

2. OBJECTIVES

2.1 Main Objective

To develop and implement a hybrid multi-modal deep learning framework that integrates peripheral blood smear image analysis with natural language processing of Complete Blood Count (CBC) reports to accurately identify Acute Lymphoblastic Leukemia (ALL) and differentiate it from Thalassemia (Minor), Iron Deficiency Anemia (IDA), and Sickle Cell Disease (SCD), optimized for Sri Lankan patient populations and healthcare infrastructure, and deployed as a web-based application to accelerate diagnosis and reduce human errors in laboratory settings.

2.2 Specific Objectives

- **Collect and preprocess Sri Lankan blood smear images and CBC reports**

To create a thorough dataset of local patient data that will be used as the basis for model training and validation, making sure that hematological changes unique to a given population are recorded in order to increase the accuracy and relevance of the diagnosis.

- **Identify and extract key morphological features of lymphoblasts**

To automatically identify and extract crucial morphological traits that are necessary for a correct classification of ALL, such as big nuclei, a high nuclear-to-cytoplasmic ratio, and irregular shapes.

- **Develop a deep learning model for lymphoblast detection and classification**

To put into practice a hybrid CNN and YOLO-based model that replicates the diagnostic procedure carried out by skilled hematologists by precisely identifying, localizing, and classifying lymphoblasts in blood smear pictures.

- **Integrate CBC parameters (WBC count, lymphoblast percentage, hemoglobin, RBC indices) into ML models**

To better differentiate ALL from other hematological disorders such Thalassemia, Iron Deficiency Anemia, and Sickle Cell Disease by analyzing supplemental quantitative data from CBC parameters in addition to morphological aspects.

- **Combine image-based and CBC-based models into a hybrid classifier**

To combine the results of CBC parameter models with morphological image analysis, using complementary data sources to improve overall diagnostic accuracy and lower false positives and negatives.

- **Validate the system with real-world clinical data**

To test the system in practical healthcare applications by piloting it for hospital-confirmed cases for the purpose of establishing clinical usability, efficacy, and reliability among different categories of patients.

- **Develop a user-friendly web-based platform for healthcare providers**

To develop a web application for laboratory staff and doctors to upload images from blood smears and CBC reports, display computer-automated diagnosis outcomes, and attain timely clinical decision-making.

3. METHODOLOGY

3.1 Overall System Description

This component focuses on the detection and identification of Acute Lymphoblastic Leukemia (ALL) through a hybrid deep learning system that combines microscopic blood smear image analysis with quantitative Complete Blood Count (CBC) parameters. The primary objective is to accurately distinguish and identify lymphoblasts while differentiating ALL from normal and other hematological disorders through computational means. To achieve this, the method employs two mutually complementary diagnostic approaches. In the image-based detection pathway, peripheral smear images are analyzed using a hybrid CNN-YOLO model, where the CNN component identifies morphological features of lymphoblasts (i.e., large nuclei, high nucleus-to-cytoplasm ratios, and atypical cell shapes), while the YOLO component localizes and detects individual lymphoblasts within each smear. In the CBC parameter analysis pathway, quantitative parameters such as WBC count, hemoglobin content, percentage of lymphoblasts, and RBC indices are processed using machine learning methods like Random Forest classifiers, providing complementary diagnostic support to distinguish ALL from other disorders with overlapping clinical features. The outputs from these two modalities are integrated through a decision-level fusion strategy that consolidates image-based and CBC-based predictions for final ALL diagnosis. Additionally, confidence scoring and uncertainty estimation mechanisms are implemented to automatically flag ambiguous cases for expert verification.

The key novelty of this approach lies in its first-of-its-kind multimodal fusion architecture that synergistically integrates visual pathological analysis with quantitative hematological CBC parameters via an advanced late fusion scheme, enhanced by intelligent uncertainty assessment that automatically identifies difficult cases requiring human expert review. This represents a significant advancement from traditional single-source diagnostic methods toward comprehensive AI-assisted medical diagnosis systems.

3.2 System Architecture Diagram

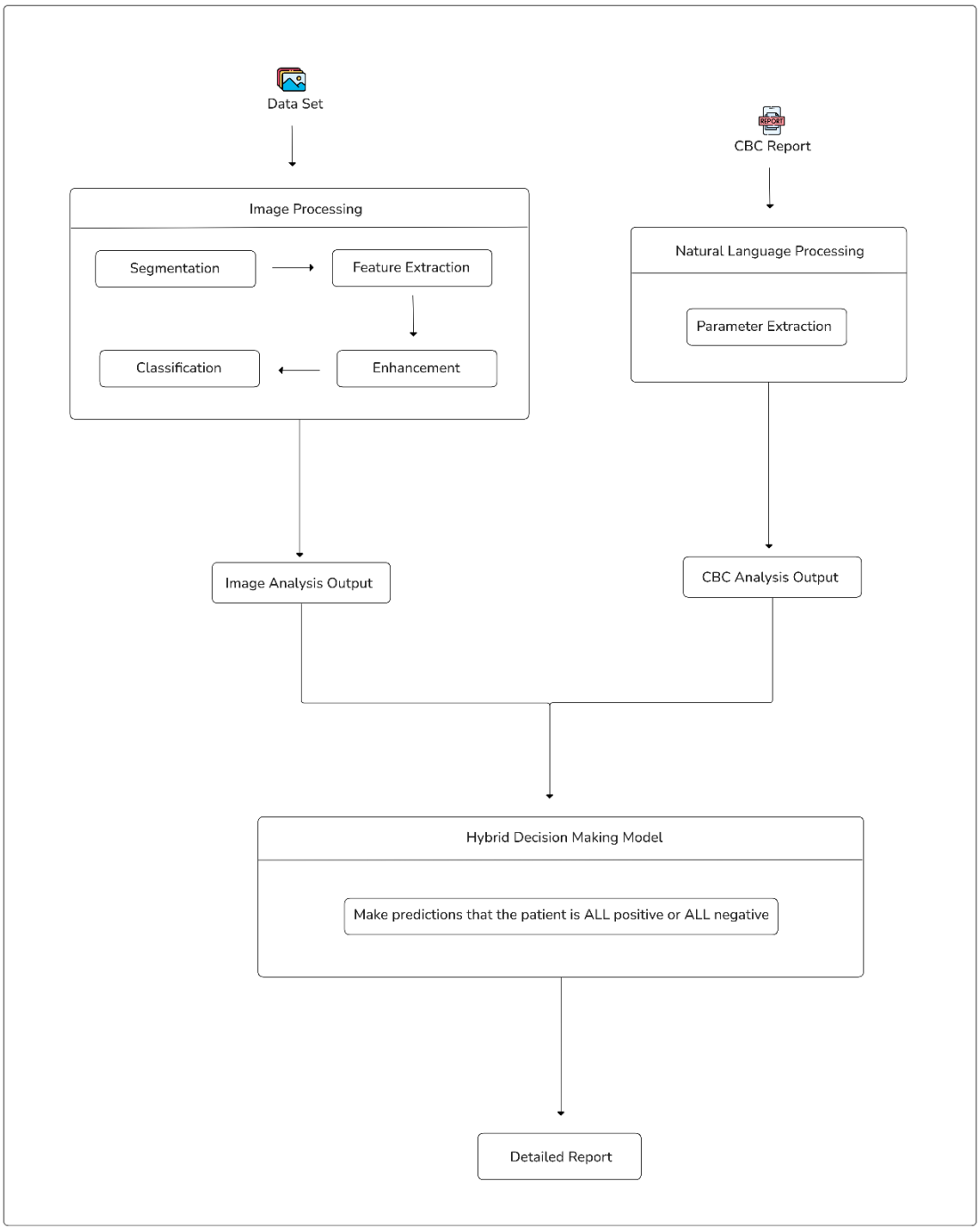

Figure 1.1: System Architecture Diagram

Figure 3.2.1 shows a hybrid medical diagnostic system that combines image analysis and clinical lab data to diagnose patients for Acute Lymphoblastic Leukemia (ALL). The system processes blood smear images through segmentation, feature extraction, enhancement, and classification steps to produce image analysis results, while

simultaneously processing CBC (Complete Blood Count) reports through natural language processing to extract key parameters. Both data streams feed into a hybrid decision-making model that makes binary predictions about whether a patient tests positive or negative for ALL, ultimately generating a detailed diagnostic report.

3.3 Identified Tasks and Sub-tasks

Task 1: Data Collection

- Sub-task 1.1: Obtain peripheral blood smear images from Sri Lankan hospitals and diagnostic centers
- Sub-task 1.2: Gather Complete Blood Count (CBC) reports from the same patients
- Sub-task 1.3: Secure ethical clearance and ensure patient anonymity

Task 2: Data Preprocessing

- Sub-task 2.1: Apply noise reduction and contrast enhancement to images
- Sub-task 2.2: Perform cell segmentation using U-Net or watershed algorithms
- Sub-task 2.3: Implement data augmentation (rotation, flipping, scaling)
- Sub-task 2.4: Normalize CBC values and organize into feature vectors
- Sub-task 2.5: Link patient blood smear features with CBC parameters

Task 3: Model Development

- Sub-task 3.1: Develop a deep learning architecture for lymphoblast morphology recognition
- Sub-task 3.2: Create machine learning classifiers for CBC parameter analysis
- Sub-task 3.3: Design hybrid fusion system combining image and CBC models

Task 4: Validation

- Sub-task 4.1: Validate system performance using real-world clinical data
- Sub-task 4.2: Calculate performance metrics (accuracy, sensitivity, specificity, precision)
- Sub-task 4.3: Perform cross validation to ensure model generalizability
- Sub-task 4.4: Compare predictions against bone marrow biopsy ground truth data

Task 5: Deployment

- Sub-task 5.1: Develop web-based platform architecture
- Sub-task 5.2: Create upload interface for blood smear images

Sub-task 5.3: Design input fields for CBC parameters

Sub-task 5.4: Implement automated diagnostic suggestions and downloadable reports

Task 6: Evaluation in Clinical Settings

Sub-task 6.1: Conduct pilot testing with Sri Lankan healthcare providers

Sub-task 6.2: Collect feedback from laboratory personnel and physicians

Sub-task 6.3: Plan integration with hospital information systems (HIS)

3.4 Materials and Resources

Hardware Resources:

- Digital microscopes or slide scanners for capturing PBS images.
- Secure data storage systems for CBC records and image datasets.

Software Resources:

- Programming languages: Python.

Human Recourses:

- Hematologists for data annotation and diagnostic validation.
- Software engineers for model development and deployment.
- Clinicians and laboratory technicians for pilot testing and usability feedback.
- System administrators for maintaining the deployed platform in hospitals.

3.5 Data Requirements and Collection

Data Needed:

- CBC Parameters: WBC count, lymphoblast percentage, hemoglobin, RBC indices
- PBS Images: High-resolution peripheral blood smear images showing lymphoblast morphologies including large nuclei, high nuclear-to-cytoplasmic ratio, and irregular cell shapes

Data Sources:

- Sri Lankan hospitals and diagnostic laboratories.

Collection Method:

- CBC data extracted from patient reports.
- PBS images are captured using digital microscopes or slide scanners.
- Annotation of PBS images by qualified hematologists to provide labels.
- CBC data and PBS images are linked securely for each patient to form a hybrid dataset.

Ethical Considerations:

- Ethical clearance obtained from relevant review committees.
- Patient consent secured where applicable.
- Data anonymized and encrypted to maintain confidentiality.

Dataset Preparation:

- Data split into training, validation, and testing sets.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- **User Authentication –** Ability of securely login to the system by healthcare professionals.
- **Data Input –** Ability to upload blood smear images and patient CBC reports**.**
- **Preprocessing -** Standardize images (resize, enhance contrast, remove noise) for analysis.
- **Disease Detection –** Ability to classify morphological features like lymphoblasts, ability to process CBC data and ability of hybrid prediction.
- **Report Generation –** Ability to provide a detailed diagnostic report including predicted disease and key indicators.
- **Data Management** – Ability to store anonymized data for further training and evaluation.

4.2 Non-Functional Requirements

- **Accuracy -** Detect Acute Lymphoblastic Leukemia (ALL) with at highest accuracy.
- **Performance -** Predictions should be made within a least amount of time.
- **Usability –** A simple, user-friendly, non-confusing interface.
- **Security and Privacy -** Patient data is anonymised and maintained securely according with medical recommendations.
- **Reliability** - Consistent and accurate results across a variety of conditions.
- **Availability -** 24 hours a day, seven days a week, with little downtime.
- **Maintainability -** Well-documented code and models that are straightforward to update.

4.3 User Requirements

The end-users of the system, which are laboratory technicians expect following.

- Ability to upload microscopic blood smear images and CBC reports into the system.
- Automatic generation of a diagnostic output that indicates whether the patient has Acute Lymphoblastic Leukemia (ALL) or not, and differentiates it from Thalassemia (Minor), Iron Deficiency Anemia (IDA), and Sickle Cell Disease (SCD).
- Easiness of the system to accessible from hospital and laboratory environments.
- Quick, reliable, and accurate results that support early diagnosis.
- Easy generation and download of a detailed report for clinical use.

4.4 System Requirements

4.4.1 Hardware requirements

- Microscope with camera attachment
- Computer with GPU (e.g., NVIDIA RTX 2060 or higher)
- Minimum 16GB RAM
- Intel i7 or equivalent processor

4.4.2 Software requirements

- Operating System: Windows 10 or Ubuntu 20.04
- Programming Language: Python 3.9 or above
- Deep Learning Frameworks: TensorFlow / PyTorch.
- Image Processing Libraries: OpenCV, Scikit-image.
- Database: MongoDB / PostgreSQL
- Web Framework: Flask / Django (for backend), React / Angular (for frontend)

5. DESCRIPTION OF PERSONAL AND FACILITIES

5.1 Personals

The project is being conducted by a group of four final-year undergraduate students who specialize in software engineering. Each member builds a unique component of the overall system, aligned with their experience.

Member 1 – Acute Lymphoblastic Leukemia (ALL) Component: Responsible for developing the component for the detection and identification of Acute Lymphoblastic Leukemia (ALL).

Member 2 – Beta-Thalassemia (Minor) Component: Responsible for developing the component for the detection and identification of Beta-Thalassemia (Minor)

Member 3 – Sickle Cell Disease (SCD) Component: Responsible for developing the component for the detection and identification of Sickle Cell Disease (SCD)

Member 4 – Iron Deficiency Anemia (IDA) Component: Responsible for developing the component for the detection and identification of Iron Deficiency Anemia (IDA)

The team collaborates on data preprocessing, model integration, testing, and web-based system deployment to ensure that all components are integrated into a single, unified platform.

5.2 Facilities

The research will utilize the following facilities and resources:

- **University laboratories** are fitted with high-performance computers and internet access to facilitate model training, testing, and development.
- **Software Resources:** TensorFlow, PyTorch, and Scikit-learn are open-source frameworks for deep learning and machine learning model construction.
- **Web development tools** include React.js, Node.js, and database management systems for creating the web-based platform.

- **Data Sources:** We worked with hospitals, laboratories, and publicly available datasets to acquire anonymized blood smear images and CBC results.
- **Cloud computing resources** include the use of platforms such as Google Colab Pro or AWS for GPU-accelerated model training when required.
- **Ethical and organizational support:** University supervisors provide guidance on gaining approvals and ethical clearance for clinical data collecting in Sri Lanka.

The combination of qualified individuals and accessible facilities ensures that the project is successfully finished within the academic schedule.

6. BUDGET AND BUDGET JUSTIFICATION

6.1 Budget

Table 6.1: Budget

Item	Description	Estimated Cost (LKR)
Cloud GPU/Computing	Occasional use of Colab Pro+ or similar platforms for model training	15,000
Web Deployment	Domain name registration and hosting for the system	8,000
Data Collection & Logistics	Printing, documentation, and travel for hospital visits	7,000
Miscellaneous	Stationery, internet, contingency expenses	5,000
Total		**35,000**

6.2 Budget Justification

- **Cloud GPU/Computing:** Free services have limited GPU support; therefore a minimal allocation is created for efficient training during peak experiments.
- **Web Deployment:** A domain name and basic hosting are required to test the system in a clinical setting.
- **Data Collection and Logistics:** This includes minimal transportation expenditures to hospitals/clinics, as well as the printing of consent forms and ethical clearance documentation.
- **Miscellaneous:** Includes modest expenses like stationery and internet data.

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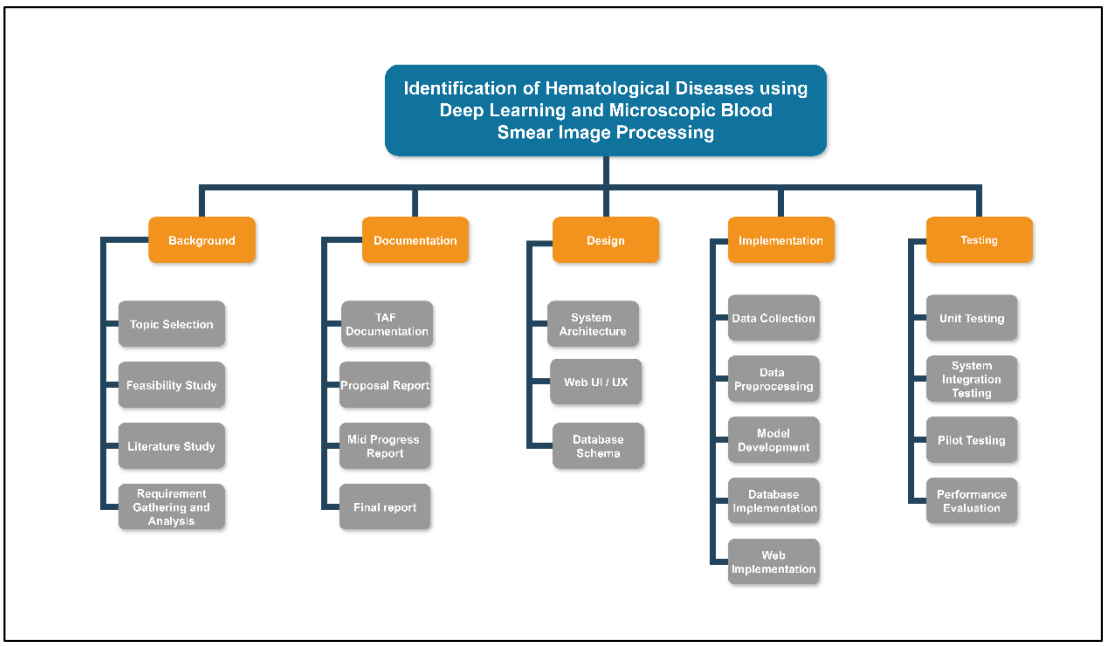
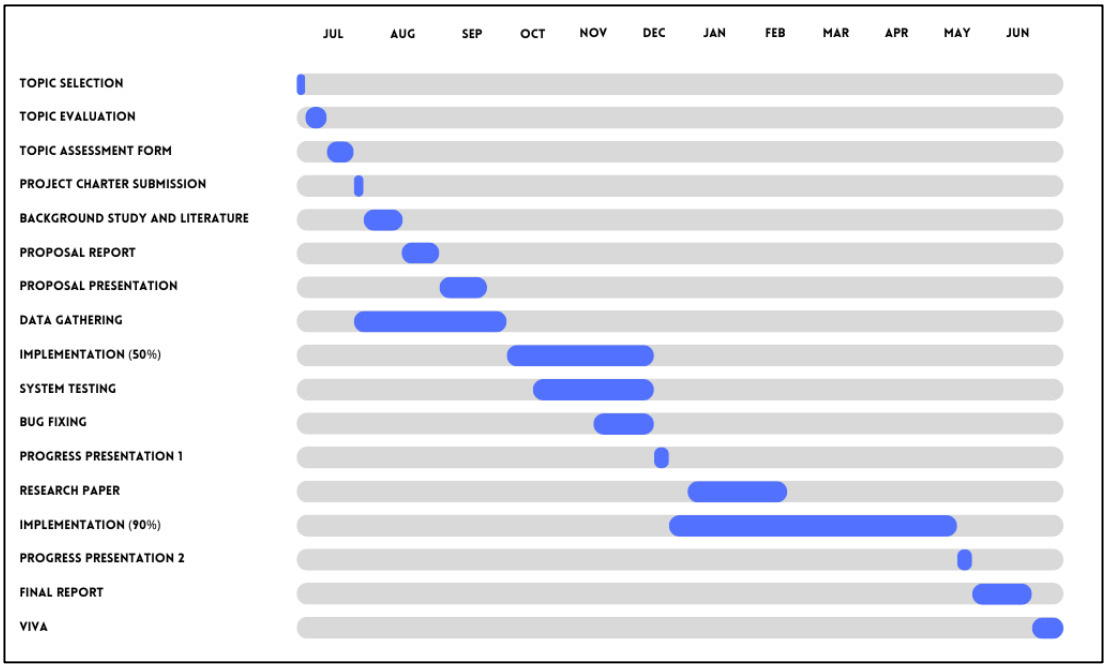
APPENDICES

Appendix A: Work Break Down Chart



Appendix B: Gantt Chart



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25-26J-514

Project Proposal Report

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August 2025

Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Hematological disorders such leukemia, beta-thalassemia, iron deficiency anemia (IDA), and sickle cell disease have a significant impact on Sri Lankan public health. Current diagnostic methods rely mainly on manual microscopic examination of blood smears, Complete Blood Count (CBC) reports, and confirmatory tests such hemoglobin electrophoresis or high-performance liquid chromatography (HPLC). While effective, these methods are resource-intensive, prone to human error, and occasionally inaccessible in rural healthcare facilities.

This research proposes a web-based intelligent diagnostic system that combines deep learning–based blood smear image analysis with machine learning models applied to CBC parameters. The methodology includes data collection from hospitals and laboratories, preprocessing of blood smear images (contrast enhancement, segmentation, augmentation), and normalization of CBC indices such as Mean Corpuscular Volume (MCV), Mean Corpuscular Hemoglobin (MCH), and RBC count. A hybrid model will then integrate both morphological features (microcytosis, hypochromia, target cells, blast cells, sickle cells) and numerical indices for accurate disease classification.

The anticipated outcome is an automated platform capable of detecting and differentiating multiple hematological diseases with higher accuracy and reliability compared to traditional manual methods. Expected test results include precise identification of Beta-Thalassemia Minor versus IDA, improved early-stage leukemia detection, and scalable classification of sickle cell abnormalities.

In conclusion, this method is expected reduce diagnostic delays, minimize human error, and improve access to reliable hematological screening in Sri Lankan hospitals and regional laboratories. Future integration with national healthcare databases, as well as adaptation for mobile health solutions, are advised to improve accessibility.

Keywords: Hematological Diseases, Deep Learning, Blood Smear Analysis, CBC Parameters, Web-based Diagnosis

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LIST OF ABBREVIATIONS

Abbreviation	Description
CBC	Complete Blood Count
MCV	Mean Corpuscular Volume
MCH	Mean Corpuscular Hemoglobin
RBC	Red Blood Cell
HPLC	High-Performance Liquid Chromatography
IDA	Iron Deficiency Anemia
DL	Deep Learning
ML	Machine Learning
AI	Artificial Intelligence
PBS	Peripheral Blood Smear

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1. INTRODUCTION

1.1 Background & Literature survey

An inherited blood disease called thalassemia is brought on by genetic disorders that can decrease or stop hemoglobin production. This genes-driven disease is one of the most common with over 200 million carriers globally [1]. Thalassemia is a serious health issue in Sri Lanka, where 4–5% of people are known to be carriers of Beta-Thalassemia Minor (trait) [2]. There is a considerable chance that two carriers will marry and have children with Beta-Thalassemia Major, a serious and life-threatening disease that requires frequent blood transfusions and intensive medical care, hence it is important to detect carriers even though they frequently remain asymptomatic [3].

A combination of peripheral blood smear analysis, complete blood count (CBC) tests, and confirmatory techniques such hemoglobin electrophoresis or high-performance liquid chromatography (HPLC) is typically used to diagnose beta-thalassemia minor [4]. While the Red Blood Cell (RBC) count is typically normal or slightly higher in CBC reports, indications like Mean Corpuscular Volume (MCV) and Mean Corpuscular Hemoglobin (MCH) are typically decreased [5]. Additional evidence is provided by peripheral blood smears because microscopic analyses frequently show target cells, hypochromia, and microcytosis, each of which are typical morphological characteristics of thalassemia [6]. Hb electrophoresis or HPLC, which identify high HbA2 levels, the gold-standard marker for thalassemia carrier status, provide the most precise confirmation [4].

Although these methods are effective, they present **several limitations**. Manual microscopic evaluation is dependent on **experienced haematologists**, is highly subjective, and often leads to variability in diagnosis [5]. More critically, thalassemia carriers can be **easily misdiagnosed as having Iron Deficiency Anemia (IDA)**, since both conditions share similar CBC findings such as low MCV and MCH [6]. Furthermore, confirmatory tests like HPLC are **expensive, time-consuming, and not readily available** in many regional healthcare facilities across Sri Lanka [3].

To solve these issues, researchers from all over the world have started using Deep Learning (DL), Machine Learning (ML), and Artificial Intelligence (AI) techniques to increase the precision and effectiveness of hematological disease diagnosis. Research has shown that red blood cell morphologies can be reliably classified using deep learning on peripheral blood smear pictures, leading to remarkable detection results [1]. Likewise, it has been demonstrated that machine learning classifiers trained on CBC data may accurately predict thalassemia carrier states; local studies conducted in Sri Lanka have claimed above 90% accuracy using models like Random Forest and Artificial Neural Networks [2][3]. More sophisticated methods, like hybrid models that include numerical and image-based data, have successfully detected thalassemia with an accuracy of 98.1% [1].

However, most of current research systems only use one kind of diagnostic input, which is either CBC indices or images. As a result, it is more difficult to identify Beta-Thalassemia Minor from IDA, which continues to be one of the most common reasons for misdiagnosis, and generalizability is reduced [6]. Furthermore, there isn't a web-based system designed especially for Sri Lankan healthcare that would enable centralized access, scalability, and effective implementation in diagnostic and medical facilities.

Recent developments have pushed AI applications beyond traditional smear and CBC analysis. For example, some studies have used deep learning on hemoglobin electrophoresis images, obtaining classification accuracies of over 95% utilizing architectures such as InceptionV3 and MobileNetV2, allowing for quick, automated thalassemia assessment from electrophoresis outputs [7]. Furthermore, emerging research has focused on real-time, point-of-care detection using AI-powered mobile technologies, such as smartphone-based applications that can screen for anemia non-invasively in the field. Such techniques show great promise for future adaptation to thalassemia carrier detection in low-resource contexts [8].

Additionally, recent reviews of **machine learning–based thalassemia prediction** emphasize the importance of considering multiple hematological parameters beyond MCV and MCH—such as RBC count, hematocrit, and RDW—to improve the

robustness and generalizability of predictive models [9]. These findings highlight the need for **hybrid systems** that integrate imaging, electrophoretic, and multivariate CBC data, which strongly supports our proposal for a Sri Lanka-specific diagnostic system.

1.2 Research Gap

Despite the significant improvements in AI-assisted hematological diagnoses, there are still a number of important gaps, especially when considering Sri Lanka:

- **Absence of Integrated Hybrid Models.**

Most AI and ML techniques now in use focused on just one type of data, such as blood smear images or numerical CBC parameters. Even if these methods individually reach a respectable level of accuracy, they fail low of taking full advantage of the valuable information that both forms of data offer. As an example, blood smear morphological alterations can reveal important qualitative information that would otherwise go unnoticed, even while CBC measurements show quantitative abnormalities [1][2][3]. However, no AI model has combined the two sources into a single framework for the detection of beta-thalassemia minor.

- **Limited Sri Lankan-specific data and models.**

Thalassemia features show ethnic and regional variation in hemoglobin indices and red blood cell morphology. Existing worldwide models, while useful elsewhere, may fail to correct for the genetic and demographic factors unique to the Sri Lankan population [2][3]. A model trained on foreign datasets may misclassify local cases or fail to detect minor geographical differences. Current Sri Lankan experiments either focus only on alpha-thalassemia [2][3] or rely on certain data types, leaving Beta-Thalassemia Minor unexplored in a hybrid AI setting.

- **Challenges in Differentiating from Similar Hematological Disorders.**

Conditions such as Iron Deficiency Anemia (IDA) often present overlapping symptoms and hematological indices with Beta-Thalassemia Minor. Existing discriminant indices and machine learning models that rely solely on CBC data

sometimes fail to achieve consistent differentiation between these conditions [4][6]. Incorporating blood smear morphology could enhance specificity, yet no hybrid approach has been tested in Sri Lanka for this purpose.

- **Practical Limitations of Current Diagnostic Approaches.**

Traditional diagnostic processes (CBC, smear analysis, and Hb electrophoresis) are time-consuming, need skilled people, and may be delayed in remote or low-resource healthcare settings. While AI can improve diagnosis, most existing technologies are either demonstrations of concept or research-focused, with little practical integration into web-based platforms suited for real-time clinical use [1][5].

In conclusion, there is surely a need for a Sri Lanka-specific hybrid diagnostic system that efficiently integrates numerical CBC data with morphological image analysis to improve the accuracy, efficiency, and clinical application of Beta-Thalassemia Minor detection.

1.3 Research Problem

According to the gaps observed, the research problem can be defined as follows:

Despite developments in AI-assisted diagnoses, present thalassemia detection techniques are limited since they rely on only CBC data or blood smear images, and most models do not fit to the Sri Lankan population [1][2][3][5]. These limitations lower diagnostic accuracy, make it difficult to identify between equivalent diseases such Iron Deficiency Anemia, and limit its practical application in clinical settings [4][6].

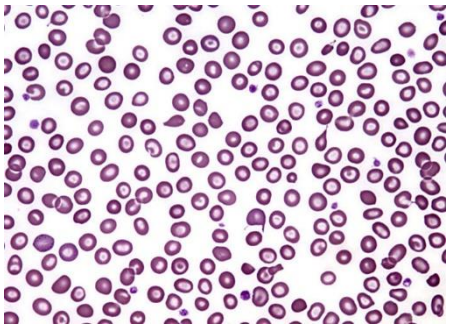

Figure 1.1: Microcytic, hypochromic red blood cells with characteristic target cells, typical of β-thalassemia [12].

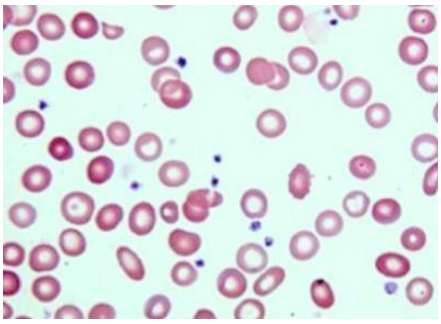

Figure 1.2:Microcytic, hypochromic red blood cells with pronounced anisopoikilocytosis (variation in size and shape), seen in moderate IDA [11].

These images *(Figure 1.1 and Figure 1.2)* visually illustrate the overlapping yet subtly different morphological patterns microcytosis and hypochromia that complicate differentiation of Thalassemia *(Figure 1.1)* and Iron Deficiency Anemia *(Figure 1.2)* by traditional AI models, reinforcing the need for your proposed population-specific diagnostic approach. [11][12]

Integration Challenge:

How may CBC numerical data and peripheral blood smear images be integrated into a single AI framework to improve diagnostic accuracy? Current models simply use pictures or CBC indices, resulting in tremendous unrealized prediction potential [1][2][3].

Sri Lankan Population-Specific Model Development:

How can a hybrid AI system be trained on Sri Lankan data to accurately record population-specific hematological variations? Models trained on foreign datasets may not accurately reflect regional genetic and environmental differences. [2][3][5].

Differential Diagnosis Capability:

How does the hybrid system differentiate between Beta-Thalassemia Minor and other diseases such as Iron Deficiency Anemia, using visual and numeric features?
Accurate differentiation is essential for preventing misdiagnosis and providing appropriate treatment care [4][6].

Web-Based System:

Developing a user-friendly web-based interface for hospitals and diagnostic labs may allow quick, scalable, and cost-effective screening. Such a system should reduce reliance on expert workers while increasing accessibility, particularly in resource-constrained healthcare settings [1][5].

To solve these challenges, a Sri Lanka-specific hybrid AI system that integrates numerical and image-based data is required, ensuring accurate detection of Beta-Thalassemia Minor, reliable differentiation from similar conditions, and practical deployment in medical environments [1][2][3][4][5][6]. This research will therefore concentrate on creating, building, and using such a system, bridging the gap between present diagnostic limitations and the potential of AI-assisted screening.

2. OBJECTIVES

2.1 Main Objectives

To accurately detect Beta-Thalassemia Minor in Sri Lankan patients, we will design, develop, and implement a hybrid, web-based diagnostic system component that combines blood smear image analysis with Complete Blood Count (CBC) characteristics. This component seeks to use cutting-edge deep learning techniques for morphological feature extraction from peripheral blood smears while also using machine learning algorithms to CBC indicators such as MCV, MCH, Hb, and RBC count. By combining these additional data sources, the component hopes to improve diagnostic accuracy, allowing genuine separation from clinically comparable disorders like Iron Deficiency Anemia, as well as developing a practical, user-friendly platform for healthcare professionals. Finally, the goal is to create a locally tailored AI system that is therapeutically useful, scalable, and accessible, reducing the need for specialized people and supporting early carrier detection to improve patient outcomes in Sri Lanka.

2.2 Specific Objectives

- **Collect and preprocess Sri Lankan blood smear images and CBC reports**

Preparing a dataset provides the foundation for training and validating models. Previous studies [2][3] have shown that incorporating local datasets enhances performance and relevance.

- **Identify and extract key morphological features of RBCs**

Microcytosis, hypochromia, and target cells are important characteristics that identify thalassemia from other anemias. Deep learning models have demonstrated high classification accuracy for certain morphological features [1][5].

- **Develop a deep learning model for RBC morphology classification**

A CNN-based model will be utilized to automatically detect aberrant cells, comparable to approaches used by hematology research [1][5].

- **Integrate CBC parameters (MCV, MCH, Hb, RBC count) into ML Models**

CBC indices are important predictors of thalassemia and can differentiate it from Iron Deficiency Anemia (IDA) [4][6]. Previous Sri Lankan investigations have confirmed that CBC indices paired with ML classifiers can achieve more than 90% accuracy [3].

- **Combine image-based and CBC-based models into a hybrid classifier**

Combined or hybrid techniques improve diagnosis accuracy by merging complementing data sources. This method has been demonstrated to improve accuracy while decreasing misclassification [1][6].

- **Validate the system with real-world clinical data.**

Testing on hospital-confirmed cases validates the system's usability and clinical relevance. Similar validation methods were used in Sri Lankan and Chinese thalassemia AI research [2][5].

- **Develop a user-friendly web-based platform for healthcare providers**

A web system enables remote access, easy image uploading, and report production. While international research prototypes exist [5][6], no Sri Lanka-specific deployment has yet been tried.

3. METHODOLOGY

3.1. Overview

The proposed methodology to identify Beta-Thalassemia Minor breaks down into multiple stages: data collection, preprocessing, developing a model, validation, and deployment. Each phase has been planned to assure accuracy, clinical relevance, and scalability in the Sri Lankan healthcare system. The methodology combines both blood smear image analysis and CBC parameter evaluation to ensure comprehensive diagnosis. Furthermore, the hybrid approach is designed to minimize misdiagnosis with conditions such as Iron Deficiency Anemia (IDA) while providing a practical web-based solution for clinical use. The step-by-step workflow of this methodology is illustrated in *Figure 3.1*.

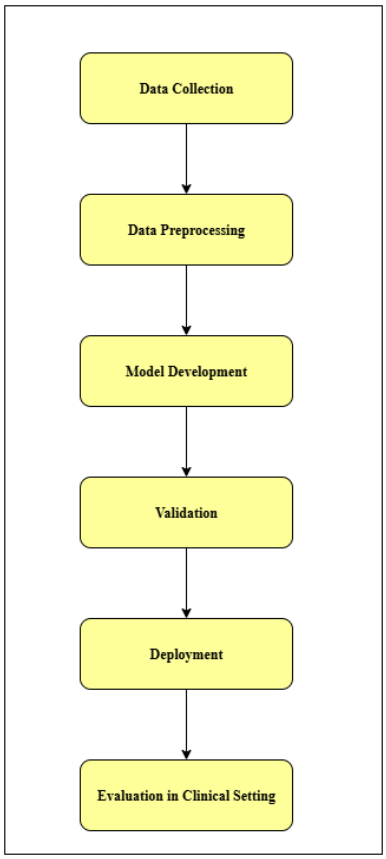

Figure 3.1: Workflow of the Proposed Methodology for Beta-Thalassemia Minor Detection

3.1.1. Step 01: Data collection

- **Peripheral blood smear images** will be obtained from Sri Lankan hospitals and diagnostic centers. The images will show both normal and proven Beta-Thalassemia Minor instances.
- **Complete Blood Count (CBC) reports** will be collected from the same patients. MCV, MCH, RBC count, and hemoglobin will be extracted.
- Confirmatory reports (Hb electrophoresis or HPLC) will serve as ground truth for model training.
- **Ethical approval:** Data collection will be carried out with ethical clearance, assuring patient anonymity and accordance with Sri Lankan health standards.
- This data gathering technique is consistent with previous studies undertaken both in Sri Lanka and globally.

3.1.2. Step 02: Data preprocessing

- **Image preprocessing**
    - Noise reduction and contrast enhancement will be used to improve cell clarity.
    - U-Net or watershed segmentation will be utilized to separate individual RBCs for analysis
    - Data augmentation (rotation, flipping, and scaling) will improve dataset diversity while reducing overfitting.

- **CBC preprocessing:**
    - The CBC values will be normalized and organized into feature vectors.
    - Missing values will be imputed using statistical techniques.

- **Integration:** Each patient's smear image features and CBC parameters will be linked to form a combined dataset for hybrid modeling.

3.1.3. Step 03: Model development

- **Image based model**
    - A deep learning architecture will be trained on RBC morphologies to identify microcytosis, hypochromia, and target cells.

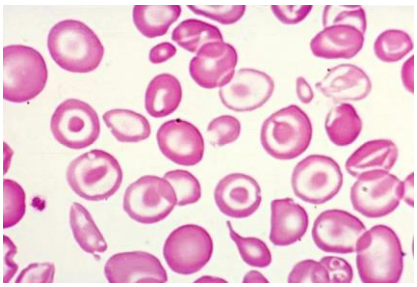

Figure 3.2:Peripheral blood smear image showing microcytic, hypochromic red blood cells typical of Beta-Thalassemia Minor [10].

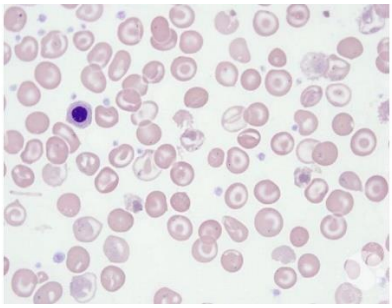

Figure 3.3:Peripheral blood smear image showing target cells, another important morphological marker in thalassemia [11].

Figure 3.2 and *Figure 3.3* highlight the key morphological features microcytosis, hypochromia, and target cells that the deep learning model will be trained to recognize. Including these representative images demonstrates how the system leverages visual data for accurate disease identification.

- **CBC based model:**
    - CBC parameters (MCV, MCH, RBC count, and Hb) will be assessed using classifiers

- **Hybrid model:**

- The outputs of the image-based and CBC-based models will be integrated using a fusion layer or ensemble technique.
- This hybrid approach is predicted to reduce misclassification, particularly when distinguishing thalassemia from Iron Deficiency Anemia (IDA)

3.1.4. Step 04: Validation

- The technology will be validated with real-world clinical data from Sri Lankan hospitals.
- Performance measurements include accuracy, sensitivity, specificity and precision,
- Cross-validation will be used to ensure model generalizability.
- The system's predictions will be compared against electrophoresis data

3.1.5. Step 05: Deployment

- The final system will be built on a web-based platform.
- **Features of the platform:**
  - Upload interface for blood smear photographs.
  - Input fields for CBC parameters.
  - Automated diagnostic suggestions (Thalassemia Trait, Normal, or Possible IDA).
  - Downloadable patient reports.
- The platform will provide remote access to laboratory technicians and clinicians, making it scalable throughout Sri Lanka.
- Web deployment ensures usability even in regional institutions that lack advanced diagnostic capabilities

3.1.6. Step 06: Evaluation in clinical settings

- Pilot testing will be conducted in collaboration with Sri Lankan healthcare providers.

- Laboratory personnel and physicians will submit comments to assist optimize the user interface and increase clinical adoption.
- The long-term goal is to integrate the platform into hospital information systems (HIS), ensuring its long-term use.

3.2. System Diagram

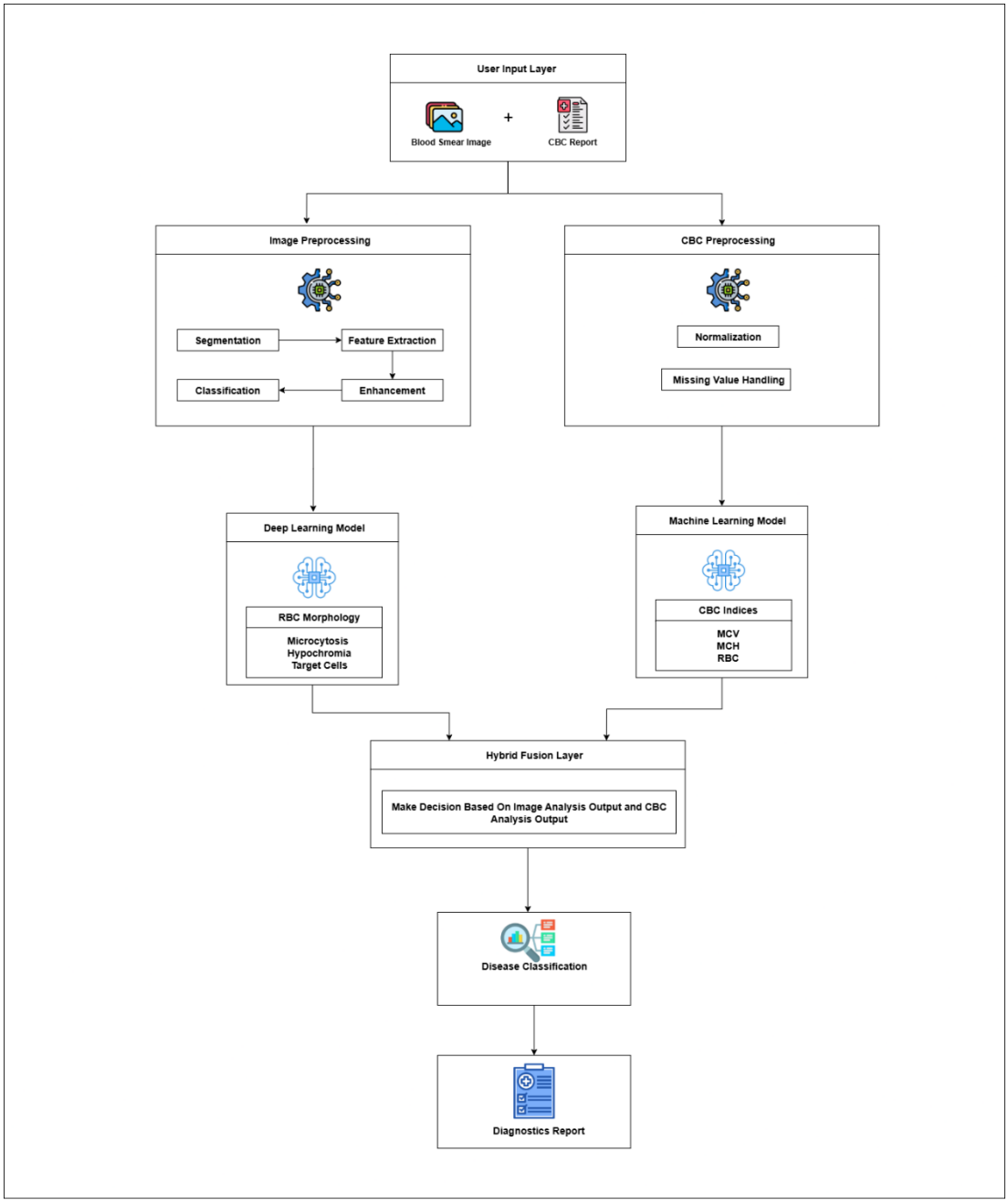

Figure 3.4: System architecture of the Beta-Thalassemia Minor detection module

Figure 3.4 shows the system architecture designed for automated detection of Beta-Thalassemia Minor. The procedure begins with the User Input Layer, which accepts

both a blood smear image and a CBC (Complete Blood Count) report. These inputs go through two independent processing processes. The Image Preprocessing module uses segmentation, enhancement, and feature extraction to emphasize and separate red blood cell properties. A Deep Learning Model analyzes these preprocessed images, identifying morphological markers such as microcytosis, hypochromia, and target cells. In parallel, the CBC Preprocessing module normalizes and handles missing values to generate numerical hematological indices such as Mean Corpuscular Volume (MCV), Mean Corpuscular Hemoglobin (MCH), and RBC count. These processed parameters are analyzed using a Machine Learning Model to identify patterns linked with Thalassemia. The outputs of both pipelines are then integrated into a Hybrid Fusion Layer, which combines image and CBC results to improve decision-making. Finally, the system performs Disease Classification and produces a thorough Diagnostic Report, giving healthcare workers a dependable, web-based tool for accurately identifying Beta-Thalassemia Minor.

3.3. Identified Tasks and Sub-tasks

Task 1: Data Collection

- Sub-task 1.1: Obtain peripheral blood smear images from Sri Lankan hospitals and diagnostic centers
- Sub-task 1.2: Gather Complete Blood Count (CBC) reports from the same patients
- Sub-task 1.3: Secure ethical clearance and ensure patient anonymity

Task 2: Data Preprocessing

- Sub-task 2.1: Apply noise reduction and contrast enhancement to images
- Sub-task 2.2: Perform cell segmentation using U-Net or watershed algorithms
- Sub-task 2.3: Implement data augmentation (rotation, flipping, scaling)
- Sub-task 2.4: Normalize CBC values and organize into feature vectors
- Sub-task 2.5: Link patient blood smear features with CBC parameters

Task 3: Model Development

- Sub-task 3.1: Develop a deep learning architecture for RBC morphologies recognition
- Sub-task 3.2: Create machine learning classifiers for CBC parameter analysis
- Sub-task 3.3: Design hybrid fusion system combining image and CBC models

Task 4: Validation

- Sub-task 4.1: Validate system performance using real-world clinical data
- Sub-task 4.2: Calculate performance metrics (accuracy, sensitivity, specificity, precision)
- Sub-task 4.3: Perform cross validation to ensure model generalizability
- Sub-task 4.4: Compare predictions against electrophoresis data

Task 5: Deployment

- Sub-task 5.1: Develop web-based platform architecture
- Sub-task 5.2: Create upload interface for blood smear images
- Sub-task 5.3: Design input fields for CBC parameters
- Sub-task 5.4: Implement automated diagnostic suggestions and downloadable reports

Task 6: Evaluation in Clinical Settings

- Sub-task 6.1: Conduct pilot testing with Sri Lankan healthcare providers
- Sub-task 6.2: Collect feedback from laboratory personnel and physicians
- Sub-task 6.3: Plan integration with hospital information systems (HIS)

3.4. Material and Resources

Hardware Resources:

- Digital microscopes or slide scanners for capturing PBS images.
- Secure data storage systems for CBC records and image datasets.

Software Resources:

- Programming languages: Python.

Human Recourses:

- Haematologists for data annotation and diagnostic validation.
- Software engineers for model development and deployment.
- Clinicians and laboratory technicians for pilot testing and usability feedback.
- System administrators for maintaining the deployed platform in hospitals.

3.5. Data requirements and collection

Data Needed:

- CBC Parameters: Hemoglobin (Hb), Mean Corpuscular Volume (MCV), Mean Corpuscular Hemoglobin (MCH), Red Blood Cell (RBC) count.
- Peripheral Blood Smear (PBS) Images: Stained blood smear images highlighting characteristic red blood cell morphologies of Beta-Thalassemia Minor, such as **microcytosis, hypochromia, target cells, and mild anisopoikilocytosis**.

Data Sources:

- Sri Lankan hospitals and diagnostic laboratories

Collection Method:

- CBC data extracted from patient reports.
- PBS images are captured using digital microscopes or slide scanners.
- Annotation of PBS images by qualified hematologists to provide labels.

- CBC data and PBS images are linked securely for each patient to form a hybrid dataset

Ethical Considerations:

- Ethical clearance obtained from relevant review committees.
- Patient consent secured where applicable.
- Data anonymized and encrypted to maintain confidentiality.

Dataset Preparation:

- Data split into training, validation, and testing sets.

4. PROJECT REQUIREMENTS

4.1 User Requirements

The end-users of the system are **laboratory technicians, pathologists, and healthcare providers** in Sri Lanka. Their key requirements are:

- Ability to upload microscopic **blood smear images** and **CBC reports** into the system.
- Automatic generation of a diagnostic output that indicates whether the patient has **Beta-Thalassemia Minor** or not, and differentiates it from conditions such as **Iron Deficiency Anemia (IDA)**.
- Easy-to-use **web interface** accessible from hospital and laboratory environments.
- Quick, reliable, and accurate results that support early diagnosis.
- Ability to generate and download a **detailed report** for clinical use.

4.2 Functional Requirements

The system must provide the following functionalities:

- **User Authentication** – Allow secure login for healthcare professionals.
- **Data Input**
    - Upload blood smear images.
    - Enter CBC parameters (MCV, MCH, RBC count, etc.).
- **Preprocessing** – Standardize images (resize, enhance contrast, remove noise) for analysis.
- **Disease Detection** –
    - Use deep learning to classify RBC morphological features (microcytosis, hypochromia, target cells).
    - Apply machine learning models on CBC data.
    - Combine both outputs for hybrid prediction.

- **Report Generation** – Provide a detailed diagnostic report including predicted disease and key indicators.
- **Data Management** – Store anonymized data for further training and evaluation.
- **System Feedback** – Allow healthcare professionals to validate or give feedback on predictions.

4.3 Non-Functional Requirements

The system must also satisfy the following quality-related requirements:

- **Accuracy:** The model should detect Beta-Thalassemia Minor with at highest accuracy
- **Performance:** Predictions should be made within a least amount of time.
- **Usability:** A simple, user-friendly interface for non-technical medical personnel.
- **Security and Privacy:** Patient data is anonymised and maintained securely according with medical recommendations.
- **Reliability:** Consistent and accurate results across a variety of conditions.
- **Availability:** 24 hours a day, seven days a week, with little downtime.
- **Maintainability** refers to well-documented code and models that are straightforward to update.

4.4 System Requirements

4.4.1 Hardware requirements

- Microscope with camera attachment
- Computer with GPU (e.g., NVIDIA RTX 2060 or higher)
- Minimum 16GB RAM
- Intel i7 or equivalent processor

4.4.2 Software requirements

- Operating System: Windows 10 or Ubuntu 20.04
- Programming Language: Python 3.9 or above
- Deep Learning Frameworks: TensorFlow / PyTorch.
- Image Processing Libraries: OpenCV, Scikit-image.
- Database: MongoDB / PostgreSQL
- Web Framework: Flask / Django (for backend), React / Angular (for frontend)

5. DESCRIPTION OF PERSONAL AND FACILITIES

5.1 Personals

The project is being conducted by a group of four final-year undergraduate students who specialize in software engineering. Each member builds a unique component of the overall system, aligned with their experience.

Member 1 – Acute Lymphoblastic Leukemia (ALL) Component: Responsible for developing the component for the detection and identification of Acute Lymphoblastic Leukemia (ALL).

Member 2 – Beta-Thalassemia (Minor) Component: Responsible for developing the component for the detection and identification of Beta-Thalassemia (Minor)

Member 3 – Sickle Cell Disease (SCD) Component: Responsible for developing the component for the detection and identification of Sickle Cell Disease (SCD)

Member 4 – Iron Deficiency Anemia (IDA) Component: Responsible for developing the component for the detection and identification of Iron Deficiency Anemia (IDA)

The team collaborates on data preprocessing, model integration, testing, and web-based system deployment to ensure that all components are integrated into a single, unified platform.

5.2 Facilities

The research will utilize the following facilities and resources:

- **University laboratories** are fitted with high-performance computers and internet access to facilitate model training, testing, and development.
- **Software Resources:** TensorFlow, PyTorch, and Scikit-learn are open-source frameworks for deep learning and machine learning model construction.

- **Web development tools** include React.js, Node.js, and database management systems for creating the web-based platform.
- **Data Sources:** We worked with hospitals, laboratories, and publicly available datasets to acquire anonymized blood smear images and CBC results.
- **Cloud computing resources** include the use of platforms such as Google Colab Pro or AWS for GPU-accelerated model training when required.
- **Ethical and organizational support:** University supervisors provide guidance on gaining approvals and ethical clearance for clinical data collecting in Sri Lanka.

The combination of qualified individuals and accessible facilities ensures that the project is successfully finished within the academic schedule.

6. BUDGET AND BUDGET JUSTIFICATION

6.1 Budget

Table 6.1: Budget Calculation

Item	Description	Estimated Cost (LKR)
Cloud GPU/Computing	Occasional use of Colab Pro+ or similar platforms for model training	15,000
Web Deployment	Domain name registration and hosting for the system	8,000
Data Collection & Logistics	Printing, documentation, and travel for hospital visits	7,000
Miscellaneous	Stationery, internet, contingency expenses	5,000
Total		**35,000**

6.2 Budget Justification

- **Cloud GPU/Computing:** Free services have limited GPU support; therefore, a minimal allocation is created for efficient training during peak experiments.
- **Web Deployment:** A domain name and basic hosting are required to test the system in a clinical setting.
- **Data Collection and Logistics:** This includes minimal transportation expenditures to hospitals/clinics, as well as the printing of consent forms and ethical clearance documentation.
- **Miscellaneous:** Includes modest expenses like stationery and internet data.

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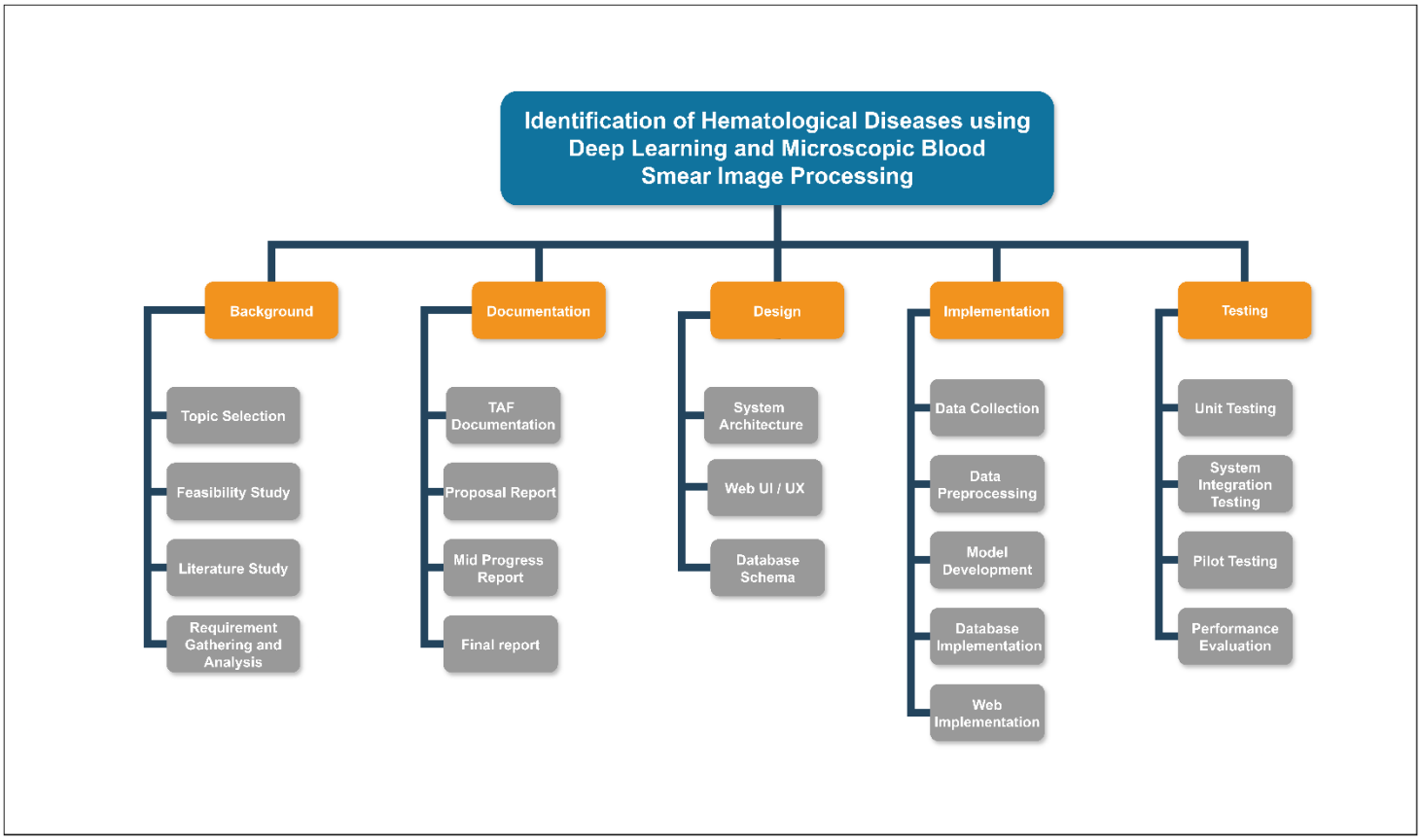
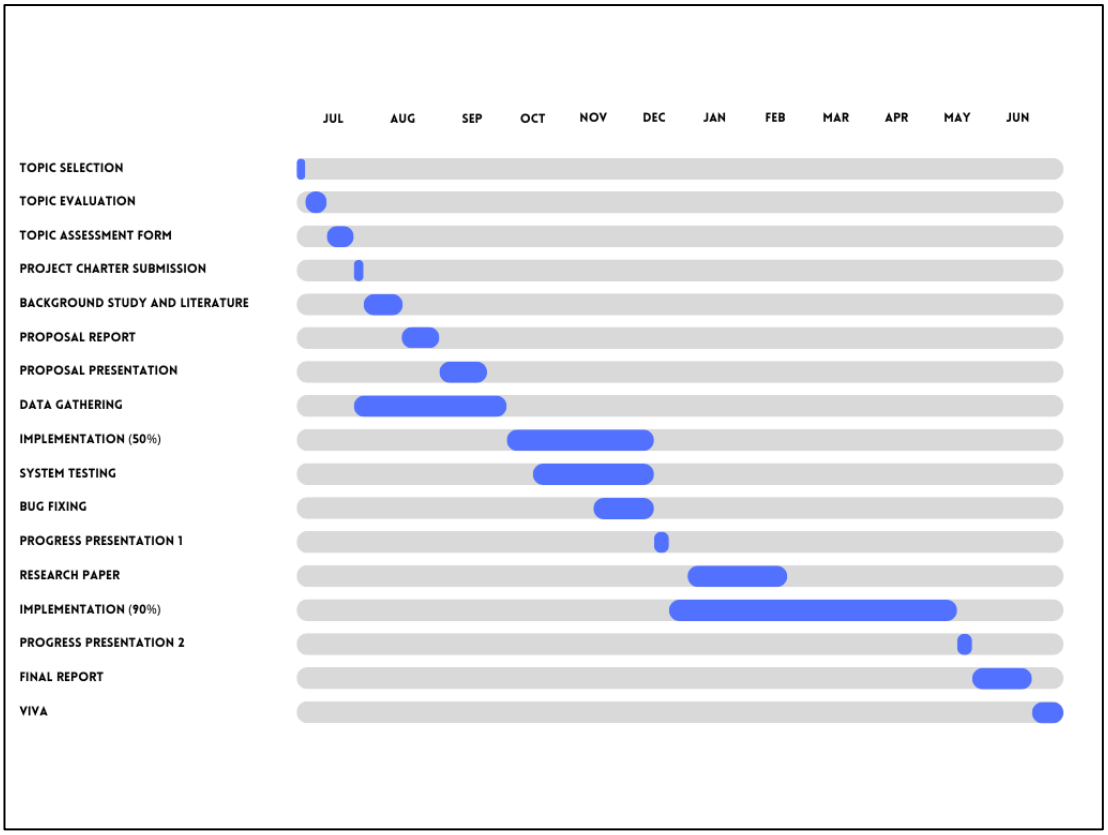
APPENDICES

Appendix A: Work Breakdown Chart



Appendix B: Gantt Chart



HEMORA – IDENTIFICATION OF HEMATOLOGICAL DISEASES USING DEEP LEARNING AND MICROSCOPIC BLOOD SMEAR IMAGE PROCESSING

25-26J-514

PROJECT PROPOSAL REPORT

Dassanayake E.G.N.L

B.Sc. (Hons) Degree in Information Technology specializing in Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology

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Sri Lanka

August 2025

DECLARATION

Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Dassanayake E.G.N.L	IT22361936	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor:
(Dr. Dinuka Wijendra)

29/08/2025
Date:

Signature of the Supervisor:
(Ms. Jenny Krishara)

29/08/2025
Date:

ABSTRACT

This proposal outlines the development of an automated detection module for Sickle Cell Disease (SCD) as a specialized component of the broader HEMORA hematological disease identification system. SCD, a genetically inherited disorder with a notable but variable prevalence in certain ethnic groups in Sri Lanka, is characterized by the deformation of red blood cells (RBCs) into a sickle shape under hypoxic conditions.[1] This morphological change leads to severe vaso-occlusive crises, chronic anemia, and progressive organ damage if not diagnosed and managed early.

The proposed solution addresses critical limitations in traditional diagnostic methods, which primarily rely on manual microscopic examination and Complete Blood Count (CBC) reports. These methods are time-consuming, subjective, and prone to human error, particularly in low-resource clinical settings where a shortage of specialized hematologists and laboratory technicians is a significant challenge .

To overcome these challenges, the project will employ a novel hybrid deep learning model. This model will integrate sophisticated image analysis of peripheral blood smear images, utilizing architectures like R-CNN for object detection and YOLO for real-time segmentation, with key CBC parameters such as Hemoglobin, Mean Corpuscular Volume (MCV), and RBC count. This multi-modal approach is designed to significantly enhance diagnostic accuracy and efficiency by combining visual and numerical data.

The methodology involves a structured approach encompassing the collection of localized clinical data from Sri Lankan patients, a comprehensive image preprocessing and segmentation pipeline, the development of the hybrid deep learning framework, and the implementation of a user-friendly, web-based interface. Anticipated outcomes include a validated system with high diagnostic accuracy, capable of reducing diagnosis time from hours to minutes, thereby alleviating healthcare burdens and facilitating more timely patient management in Sri Lankan clinical environments.

Keywords: Sickle Cell Disease, Deep Learning, Blood Smear Images, CBC Data, Hybrid Model.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CBC	Complete Blood Count
CNN	Convolutional Neural Network
Hb	Hemoglobin
MCV	Mean Corpuscular Volume
RBC	Red Blood Cell
R-CNN	Region-based Convolutional Neural Network
SCD	Sickle Cell Disease
SDLC	Software Development Life Cycle
ViT	Vision Transformer
WBS	Work Breakdown Structure
YOLO	You Only Look Once

1. INTRODUCTION

1.1 Background & Literature Survey

The field of medicine, particularly hematological diagnostics, is undergoing a profound transformation driven by advancements in Artificial Intelligence (AI) and deep learning. Hematological diseases, which affect the blood and the blood-forming organs, represent a significant public health challenge in Sri Lanka and globally. Conditions such as Leukemia, Thalassemia, Iron Deficiency Anemia (IDA), and Sickle Cell Disease (SCD) can lead to severe health complications, including developmental delays, organ damage, and even death if not diagnosed and treated promptly.

Traditional diagnostic methodologies in Sri Lanka face substantial systemic challenges. A significant obstacle is the heavy reliance on manual microscopic examination of peripheral blood smear slides. This method is inherently labor-intensive, time-consuming, and highly susceptible to inter-observer variability, which can lead to inconsistencies in diagnosis . The situation is further compounded by a shortage of specialized hematologists and laboratory technicians, particularly in rural regions of the country. This unequal distribution of healthcare resources necessitates that many patients travel long distances for proper diagnostic tests, leading to significant delays in receiving crucial diagnoses and initiating treatment . In many government hospitals, routine blood test results can face delays of 3 to 7 days, a timeframe that can be critical in managing aggressive conditions like leukemia . The development of an automated, accurate, and accessible diagnostic tool is therefore not merely an enhancement but a critical necessity to overcome these systemic limitations and improve patient outcomes.

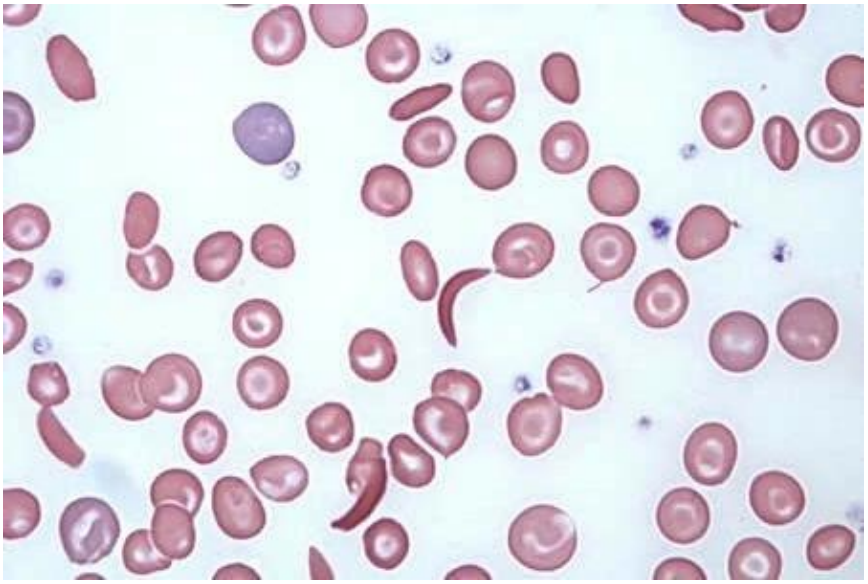

Figure 1: Sickle Shaped RBC of an Affected Patient

Within this broader context, the identification of Sickle Cell Disease presents a specific set of challenges. SCD is a genetically inherited disorder caused by a mutation in the gene encoding hemoglobin. This mutation leads to the production of abnormal hemoglobin (HbS) which, under low oxygen conditions, causes red blood cells (RBCs) to polymerize and assume a rigid, sickle-like shape.[1] While the overall prevalence of SCD in Sri Lanka is relatively low, around 2.8% in screened populations, it exhibits a milder Asian phenotype and its incidence varies significantly among different ethnic groups, such as the Sinhalese population . This nuance makes manual detection particularly difficult for less experienced medical personnel, increasing the risk of delayed or inaccurate diagnoses. The manual process involves not only identifying the distinctive sickle-shaped cells under a microscope but also corroborating these visual findings with numerical data from Complete Blood Count (CBC) reports, such as Hemoglobin levels, MCV, and RBC count.

Recent advancements in deep learning have shown great promise in automating and improving the accuracy of hematological diagnostics. These technologies can process and analyze complex visual data from microscopic images with a speed and consistency that surpasses human capabilities. Convolutional Neural Networks (CNNs), in particular, have demonstrated excellent performance in image classification and feature extraction from medical images. Deep learning models, when trained on annotated datasets, can automatically identify specific abnormalities, such as blast cells in leukemia or hypochromic microcytic cells in anemia, with high accuracy in a short period.

The current body of literature on AI for hematological disease detection provides a solid foundation but also highlights specific areas that require further exploration. A significant portion of existing research has focused on pure image-based deep learning models. For instance, studies by Alzubaidi et al. (2020) demonstrated the efficacy of deep learning for classifying RBCs from microscopy images . De Haan et al. (2020) specifically explored a smartphone-based microscope system with an integrated deep learning model, showing its potential for automated screening in low-resource settings . These works establish the feasibility and utility of applying deep learning to blood smear image analysis.

However, the field is evolving toward more sophisticated, multi-modal approaches. The single-input, image-based methods, while effective, do not leverage all available diagnostic information. The integration of various data sources, such as numerical CBC parameters, is gaining traction. Purwar et al. (2020) used CNNs to extract features from both CBC reports and blood films to detect microcytic hypochromia, a condition relevant to SCD, demonstrating the value of combining visual and numerical data for enhanced diagnostic reliability .

Despite these developments, a critical need remains for solutions that are specifically tailored to the unique epidemiological and clinical contexts of different regions. Most publicly available datasets for hematological diseases originate from high-prevalence areas like Africa or the US, and their models may not generalize effectively to the specific disease variants found elsewhere, such as the milder Asian phenotype of SCD prevalent in Sri Lanka . Furthermore, while some research explores the technical feasibility of AI-driven diagnostics, there is a distinct lack of solutions that address the practical deployment challenges in low-resource contexts by providing user-friendly, web-based interfaces for remote access. The present research is designed to address these specific and critical shortcomings by creating a localized, hybrid, and deployable solution for the detection of Sickle Cell Disease.

1.2 Research Gap

While the literature has demonstrated the significant potential of deep learning for hematological disease detection, a thorough analysis reveals several critical gaps that limit the generalizability, accuracy, and practical utility of existing solutions. These gaps are particularly pronounced when considering the specific context of the Sri Lankan healthcare system and the unique characteristics of the disease variant prevalent in the region.

First, there is a fundamental gap in data integration. The majority of published research on SCD detection focuses almost exclusively on image-based deep learning models. While these models are highly effective at identifying the morphological features of sickle-shaped cells, they often do not incorporate crucial supporting diagnostic information from Complete Blood Count (CBC) reports. The isolated use of image analysis may lead to incomplete diagnostic decisions, potentially confusing SCD with other conditions that have similar, though not identical, visual markers. A more comprehensive approach requires a hybrid framework that can simultaneously analyze the visual characteristics of blood cells and the numerical hematological parameters that are essential for a complete and reliable diagnosis. The absence of such a holistic model in much of the existing literature represents a significant technical deficit .

Second, a major deficiency exists in the localization and adaptation of existing datasets and models. Most deep learning models for hematological diagnostics are trained on datasets sourced from high-prevalence regions, such as parts of Africa or the United States, which have a different disease prevalence and potentially a different phenotypic presentation of SCD. The "milder Asian phenotype" of SCD found in Sri Lanka may not be adequately represented in these global datasets, leading to models that perform sub-optimally when applied to the local patient population . The lack of a comprehensive, annotated dataset of blood smear images and CBC reports from Sri Lankan patients is a critical barrier to developing models

that are clinically sensitive and specific for the local context . Creating a model that is trained and validated on local data is a direct way to bridge this gap and ensure diagnostic reliability in the target environment.

Third, a significant gap in the practical deployment and accessibility of these systems has been identified. While some research has explored the use of AI in low-resource settings, few of these studies have resulted in a robust, user-friendly, and web-based solution that is easily accessible to healthcare professionals. A practical system must not only be technically sound but also be compatible with existing clinical workflows. The reliance on specialized hardware or complex software setups can prevent widespread adoption. A web-based interface that allows for remote data upload, fast processing, and report generation is a crucial requirement for making this technology a viable and scalable solution in a country with significant urban-rural disparities in healthcare infrastructure. The current absence of such a practical and accessible solution limits the real-world impact of AI in this domain .

Finally, the existing literature shows a need for more effective and efficient model architectures. While deep learning models like CNNs and R-CNNs have been used, there is an opportunity to improve both the accuracy and efficiency of detection through a more advanced hybrid approach. The combination of algorithms like R-CNN, known for its high-accuracy object detection, and YOLO, celebrated for its real-time performance, has not been thoroughly explored in the context of SCD diagnostics. A new model that effectively integrates these complementary strengths could lead to a system that is both highly accurate and fast, which is essential for a clinical tool. The novelty of this project lies in creating a hybrid model that fuses visual features from a combined R-CNN/YOLO architecture with numerical CBC data, a unique combination specifically designed to address the aforementioned gaps and provide a more comprehensive, accurate, and practical diagnostic solution.

1.3 Research Problem

The manual identification of Sickle Cell Disease (SCD) in Sri Lanka presents a complex challenge, characterized by a confluence of technical, logistical, and human factors. The process relies on the microscopic analysis of peripheral blood smears to detect abnormal, sickle-shaped Red Blood Cells (RBCs) and the concurrent evaluation of Complete Blood Count (CBC) metrics, such as low Hemoglobin (Hb), MCV, and RBC count. However, this traditional approach is inherently flawed. It is a time-consuming and laborious process that places a significant burden on laboratory staff and clinical personnel .

The problem is further aggravated by the relatively low but ethnically variable prevalence of SCD in the country, which means that laboratory technicians may not have frequent exposure to the condition. This infrequent encounter can increase the

likelihood of delayed or inaccurate diagnoses, especially in under-equipped clinics where access to experienced hematologists is limited . A misdiagnosis or a late diagnosis can have severe consequences for a patient's quality of life and treatment outcomes.

The current diagnostic ecosystem in Sri Lanka is thus confronted with a critical question: How can deep learning be leveraged to create an automated and reliable system for SCD detection that not only improves diagnostic accuracy and efficiency but is also tailored to the specific needs of local clinical settings? The solution must be capable of overcoming the subjectivity and delays associated with manual analysis by integrating the complementary strengths of image-based morphology and numerical hematological parameters. This project aims to directly address this multifaceted problem by developing a robust, hybrid deep learning solution designed for the Sri Lankan context.

2. OBJECTIVES

The overarching goal of this research is to apply advanced deep learning and image processing techniques to the critical domain of hematological diagnostics. This project's specific focus is on creating a specialized, automated module for the detection of Sickle Cell Disease. The module will be a foundational component of a larger system, the HEMORA project, which seeks to identify multiple hematological disorders with high precision and efficiency. The primary objective is to create a diagnostic tool that is not only technically advanced but also practically applicable in a real-world clinical environment, particularly in Sri Lanka's low-resource settings, thereby supporting medical personnel in making accurate and timely decisions.

2.1 Main Objective

The main objective is to develop an automated module for the detection and identification of Sickle Cell Disease (SCD) within a comprehensive hematological disease identification system. This module will integrate deep learning-based analysis of microscopic blood smear images with Complete Blood Count (CBC) parameters to achieve accurate, timely diagnoses, supporting medical personnel in Sri Lanka's low-resource environments. The system's successful implementation will provide a scalable and reliable tool that reduces diagnostic delays and improves the quality of care for patients with hematological conditions.

2.2 Specific Objectives

The following specific objectives have been established to guide the development and implementation of the proposed SCD detection module. These objectives are designed to be measurable and to follow a logical progression, from data acquisition to final deployment.

1. **Data Collection and Curation:** To collect and curate a unique, localized dataset by September 2025. This dataset will consist of peripheral blood smear images and corresponding CBC reports from SCD cases in Sri Lanka, which is essential for adapting the model to the specific "milder Asian phenotype" prevalent in the region. The collection process will ensure ethical approvals and patient data anonymization.
2. **Image Preprocessing and Segmentation:** To preprocess and segment individual red blood cells (RBCs) from the blood smear images using advanced computer vision techniques by October 2025. This task involves applying a multi-stage pipeline that includes resizing, contrast enhancement, and noise reduction, followed by accurate cell isolation using algorithms such as thresholding and the watershed algorithm to prepare the data for efficient deep learning analysis.
3. **Hybrid Deep Learning Model Development:** To develop a novel hybrid deep learning model by December 2025. The model will combine the strengths of object detection algorithms like R-CNN and YOLO to accurately identify sickle-shaped cells and other morphological signs in the segmented RBCs. The model will be fine-tuned to achieve a high degree of accuracy, with a target of at least 90% on test data.
4. **Data Fusion Framework:** To create a dedicated data fusion framework to integrate numerical CBC parameters (Hemoglobin, MCV, and RBC count) with the visual features extracted by the deep learning model by January 2026. This step is crucial for enhancing the model's decision-making process, allowing it to make more reliable and comprehensive diagnoses by considering both morphological and physiological data.
5. **Model Training and Evaluation:** To train and evaluate the combined hybrid model on the curated local datasets by February 2026. This process will involve rigorous performance measurement using standard metrics such as precision, recall, F1-score, and Area Under Curve (AUC) to ensure the model's robustness and clinical validity.
6. **Clinical Validation:** To validate the model's output with real patient data from clinical partners to ensure its practical applicability and reliability in a hospital environment by March 2026. This crucial step will bridge the gap between a theoretical model and a functional clinical tool.
7. **System Implementation:** To implement a secure, user-friendly, and responsive web-based interface for healthcare professionals by April 2026. This interface will allow for the seamless upload of blood smear images and CBC reports, providing a platform for easy data input and the display of diagnostic results, thereby facilitating remote access and widespread adoption in Sri Lankan laboratories.

3. METHODOLOGY

3.1 Overview

The conceptual architecture of the SCD detection module is designed to handle both the methodology for our Sickle Cell Disease (SCD) detection system is built on a hybrid approach that combines visual analysis of blood smear images with numerical data from a patient's complete blood count (CBC) report. Our process is designed to be both comprehensive and iterative, ensuring that the final system is not only highly accurate but also practical for clinical use.

The project begins with a crucial data acquisition phase, where we collect pairs of microscopic blood smear images and their corresponding CBC reports. To ensure our model is relevant to the local population, we are primarily sourcing this data from clinical partners in Sri Lanka, alongside augmenting our dataset with publicly available sources. This dual-pronged approach helps create a powerful training resource that is both tailored and comprehensive. To prepare the images for analysis, they undergo a series of preprocessing steps, including resizing, color normalization, and contrast enhancement, to ensure consistency and improve clarity. We then employ advanced computer vision techniques, such as the watershed algorithm, to accurately segment and isolate individual red blood cells. This step is vital because it allows our model to focus on the morphological features of each cell, such as its shape and size, with high precision.

The core of our system is a hybrid deep learning model that processes these two distinct data streams. For the visual analysis, we've developed a novel combination of a Region-based Convolutional Neural Network (R-CNN) and the "You Only Look Once" (YOLO) algorithm. This hybrid architecture leverages the high accuracy of R-CNN for object detection and the real-time efficiency of YOLO for cell classification, allowing us to quickly and accurately identify sickle-shaped cells. Concurrently, the numerical CBC data—including crucial metrics like hemoglobin and MCV—is processed separately. This numerical data provides essential diagnostic context that visual analysis alone can't offer, helping to differentiate SCD from other conditions with similar symptoms. The outputs from both the visual and numerical pipelines are then fused together in a dedicated Data Fusion Layer. This layer is where the system truly excels, as it learns the complex relationships between the visual appearance of the cells and the patient's hematological parameters, leading to a more reliable and holistic diagnosis.

Finally, the system undergoes a rigorous evaluation and validation process. We measure the model's performance using standard clinical metrics like precision, recall, and F1-score to ensure its effectiveness. The ultimate test, however, will be a clinical validation phase, where we will compare our system's automated diagnoses

against those of expert medical professionals using new, real-world data. This final step is crucial for building trust and demonstrating the system's potential to become a reliable diagnostic tool in a practical healthcare setting.

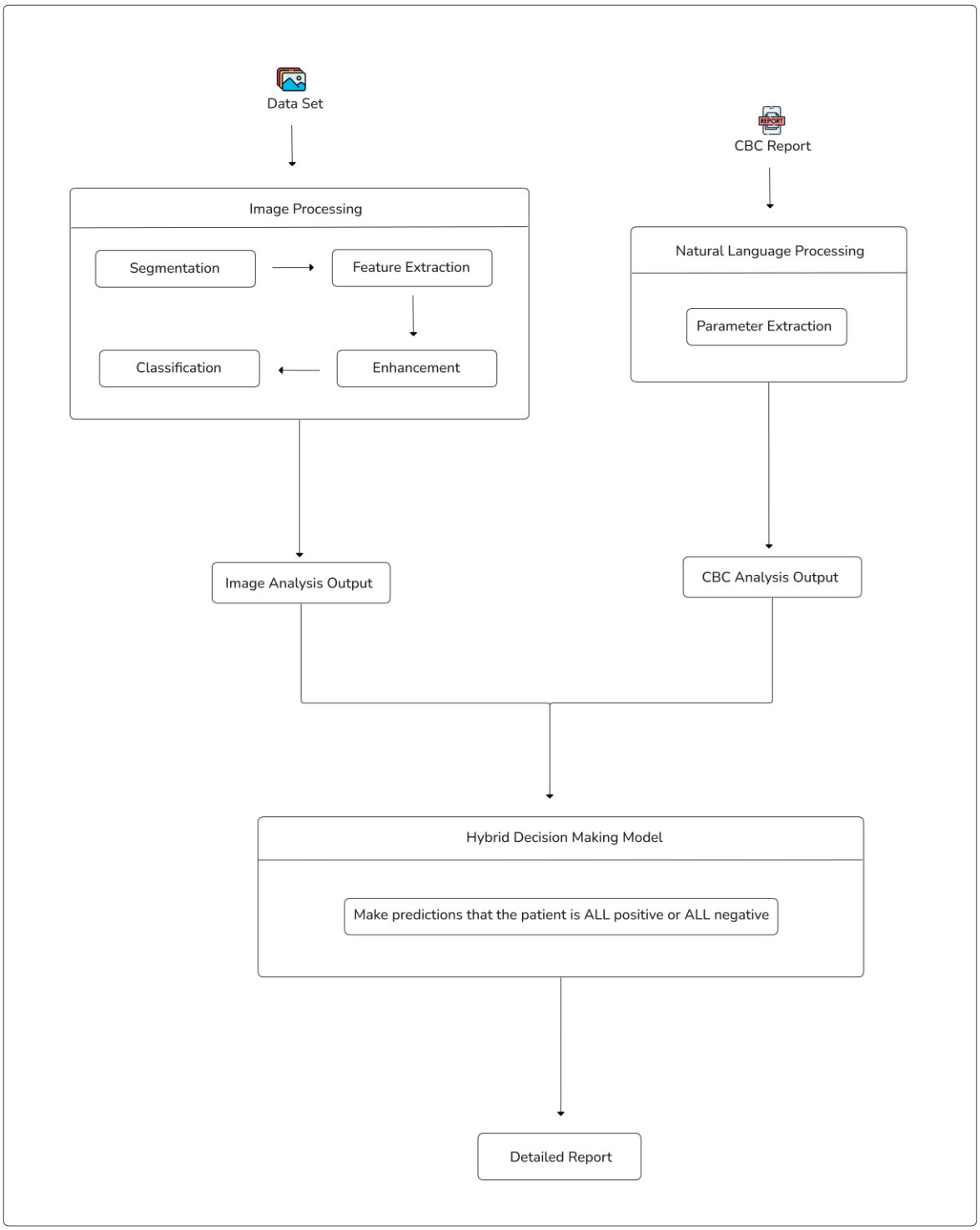

Figure 2: Expanded System Architecture for the SCD Detection Module

3.2 Identified Tasks and Sub-tasks

Task 1: Data Collection

Sub-task 1.1: Obtain peripheral blood smear images from Sri Lankan hospitals and diagnostic centers

Sub-task 1.2: Gather Complete Blood Count (CBC) reports from the same patients

Sub-task 1.3: Secure ethical clearance and ensure patient anonymity

Task 2: Data Preprocessing

Sub-task 2.1: Apply noise reduction and contrast enhancement to images

Sub-task 2.2: Perform cell segmentation using U-Net or watershed algorithms

Sub-task 2.3: Implement data augmentation (rotation, flipping, scaling)

Sub-task 2.4: Normalize CBC values and organize into feature vectors

Sub-task 2.5: Link patient blood smear features with CBC parameters

Task 3: Model Development

Sub-task 3.1: Develop a deep learning architecture for lymphoblast morphology recognition

Sub-task 3.2: Create machine learning classifiers for CBC parameter analysis

Sub-task 3.3: Design hybrid fusion system combining image and CBC models

Task 4: Validation

Sub-task 4.1: Validate system performance using real-world clinical data

Sub-task 4.2: Calculate performance metrics (accuracy, sensitivity, specificity, precision)

Sub-task 4.3: Perform cross validation to ensure model generalizability

Sub-task 4.4: Compare predictions against bone marrow biopsy ground truth data

Task 5: Deployment

Sub-task 5.1: Develop web-based platform architecture

Sub-task 5.2: Create upload interface for blood smear images

Sub-task 5.3: Design input fields for CBC parameters

Sub-task 5.4: Implement automated diagnostic suggestions and downloadable reports

Task 6: Evaluation in Clinical Settings

Sub-task 6.1: Conduct pilot testing with Sri Lankan healthcare providers

Sub-task 6.2: Collect feedback from laboratory personnel and physicians

Sub-task 6.3: Plan integration with hospital information systems (HIS)

3.3 Materials and Resources Hardware Resources:

- Digital microscopes or slide scanners for capturing PBS images.
- Secure data storage systems for CBC records and image datasets.

Software Resources:

- Programming languages: Python.

Human Recourses:

- Hematologists for data annotation and diagnostic validation.
- Software engineers for model development and deployment.
- Clinicians and laboratory technicians for pilot testing and usability feedback.
- System administrators for maintaining the deployed platform in hospitals.

3.4 Data requirements and collection

Data Needed:

- CBC Parameters: Hb, MCV, MCH, RDW
- PBS Images: Stained blood smear images showing RBC morphologies such as microcytosis, hypochromia, anisocytosis, and poikilocytosis.

Data Sources:

- Sri Lankan hospitals and diagnostic laboratories

Collection Method:

- CBC data extracted from patient reports.
- PBS images are captured using digital microscopes or slide scanners.
- Annotation of PBS images by qualified hematologists to provide labels.
- CBC data and PBS images are linked securely for each patient to form a hybrid dataset

Ethical Considerations:

- Ethical clearance obtained from relevant review committees.
- Patient consent secured where applicable.
- Data anonymized and encrypted to maintain confidentiality.

Dataset Preparation:

- Data split into training, validation, and testing sets.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

Functional requirements define what the system must do. These requirements are directly tied to the core objectives of the project and describe the specific behaviors and functionalities of the SCD detection module.

1. **Image and Data Upload**
   - The system shall allow users to upload digital microscopic blood smear images and input numerical CBC report data.
2. **RBC Segmentation**
   - The system shall accurately segment individual red blood cells (RBCs) from the background of the uploaded blood smear image.
3. **Morphological Classification**
   - The system shall classify the shape of individual RBCs, specifically identifying and counting sickle-shaped cells.
4. **Hybrid Prediction Engine**
   - The system shall integrate the analysis of image-derived features and numerical CBC parameters to generate a final diagnostic prediction.
5. **Diagnostic Report Generation**
   - The system shall generate a detailed diagnostic report that includes the SCD probability score, confidence level, and visual highlights of detected sickle cells.
6. **User Interface**
   - The system shall provide an intuitive, user-friendly, and role-based web interface for healthcare professionals to interact with the system.
7. **Model Retraining**
   - The system shall include a feature for authorized administrators to retrain the deep learning model with new, labeled data to improve its performance over time.

4.2 Non-Functional Requirements

Non-functional requirements specify the criteria for evaluating the operation of a system, rather than its specific behaviors. These requirements are crucial for ensuring the quality, usability, and security of the SCD detection module in a clinical environment.

1. **Accuracy**
   - The system's diagnostic accuracy shall be greater than 90% on the final test data, with a target F1-score of at least 0.90 for both positive and negative cases.

2. **Response Time**
   - The system shall process a single image and CBC report and provide a diagnostic result in less than 10 seconds.

3. **Scalability**
   - The system shall be scalable to handle a minimum of 100 simultaneous users without compromising performance or response time.

4. **Security**
   - The system shall ensure secure data handling and communication, including encryption of patient data and a secure authentication mechanism for users.

5. **Reliability**
   - The system shall be highly reliable, with an uptime of at least 99.5%, ensuring continuous availability for clinical use.

6. **Usability**
   - The user interface shall be intuitive and require minimal training for laboratory staff and doctors to operate effectively.

7. **Compatibility**
   - The web interface shall be compatible with modern web browsers, including Chrome, Firefox, and Edge.

4.3 User Requirements

User requirements describe what the end-users of the system need and expect. For the SCD detection module, the primary users are healthcare professionals, including laboratory staff and doctors, who will be interacting with the system directly.

8. **Data Upload**
   - Lab staff must be able to easily upload blood smear images and input CBC data via the web interface.

9. **Result Viewing**
    - Doctors must be able to view a clear and concise diagnostic report, including the SCD probability and a confidence score.

10. **Visualization**
    - Users should be able to view a visual representation of the detected cells on the blood smear image to verify the model's predictions.

11. **Role-Based Access**
    - The system shall provide different access levels for users, such as a read-only role for lab staff and an administrative role for model updates and system management.

4.4 System requirements

4.4.1 Hardware requirements

- Microscope with camera attachment
- Computer with GPU (e.g., NVIDIA RTX 2060 or higher)
- Minimum 16GB RAM
- Intel i7 or equivalent processor

4.4.2 Software requirements

- Operating System: Windows 10 or Ubuntu 20.04
- Programming Language: Python 3.9 or above
- Deep Learning Frameworks: TensorFlow / PyTorch.
- Image Processing Libraries: OpenCV, Scikit-image.
- Database: MongoDB / PostgreSQL
- Web Framework: Flask / Django (for backend), React / Angular (for frontend)

5. Description of Personal and Facilities

5.1 Personals

The project is being conducted by a group of four final-year undergraduate students who specialize in software engineering. Each member builds a unique component of the overall system, aligned with their experience.

- Member 1 - Acute Lymphoblastic Leukemia (ALL) Component: Responsible for developing the component for the detection and identification of Acute Lymphoblastic Leukemia (ALL).
- Member 2 - Beta-Thalassemia (Minor) Component: Responsible for developing the component for the detection and identification of Beta-Thalassemia (Minor)
- Member 3 - Sickle Cell Disease (SCD) Component: Responsible for developing the component for the detection and identification of Sickle Cell Disease (SCD)
- Member 4 - Iron Deficiency Anemia (IDA) Component: Responsible for developing the component for the detection and identification of Iron Deficiency Anemia (IDA)

The team collaborates on data preprocessing, model integration, testing, and web-based system deployment to ensure that all components are integrated into a single, unified platform.

5.2 Facilities

The research will utilize the following facilities and resources:

- University laboratories are fitted with high-performance computers and internet access to facilitate model training, testing, and development.

- Software Resources: TensorFlow, PyTorch, and Scikit-learn are open-source frameworks for deep learning and machine learning model construction.

- Web development tools include React.js, Node.js, and database management systems for creating the web-based platform.

- Data Sources: We worked with hospitals, laboratories, and publicly available datasets to acquire anonymized blood smear images and CBC results.

- Cloud computing resources include the use of platforms such as Google Colab Pro or AWS for GPU-accelerated model training when required.

- Ethical and organizational support: University supervisors provide guidance on gaining approvals and ethical clearance for clinical data collecting in Sri Lanka.

The combination of qualified individuals and accessible facilities ensures that the project is successfully finished within the academic schedule.

6. BUDGET AND BUDGET JUSTIFICATION

6.1 Budget

Table 1: Budget Table

Item	Description	Estimated Cost (LKR)
Cloud GPU/Computing	Occasional use of Colab Pro+ or similar platforms for model training	15,000
Web Deployment	Domain name registration and hosting for the system	8,000
Data Collection & Logistics	Printing, documentation, and travel for hospital visits	7,000
Miscellaneous	Stationery, internet, contingency expenses	5,000
Total		**35,000**

6.2 Budget Justification

- Cloud GPU/Computing: Free services have limited GPU support. therefore, a minimal allocation is created for efficient training during peak experiments.
- Web Deployment: A domain name and basic hosting are required to test the system in a clinical setting.
- Data Collection and Logistics: This includes minimal transportation expenditures to hospitals/clinics, as well as the printing of consent forms and ethical clearance documentation.
- Miscellaneous: Includes modest expenses like stationery and internet data.

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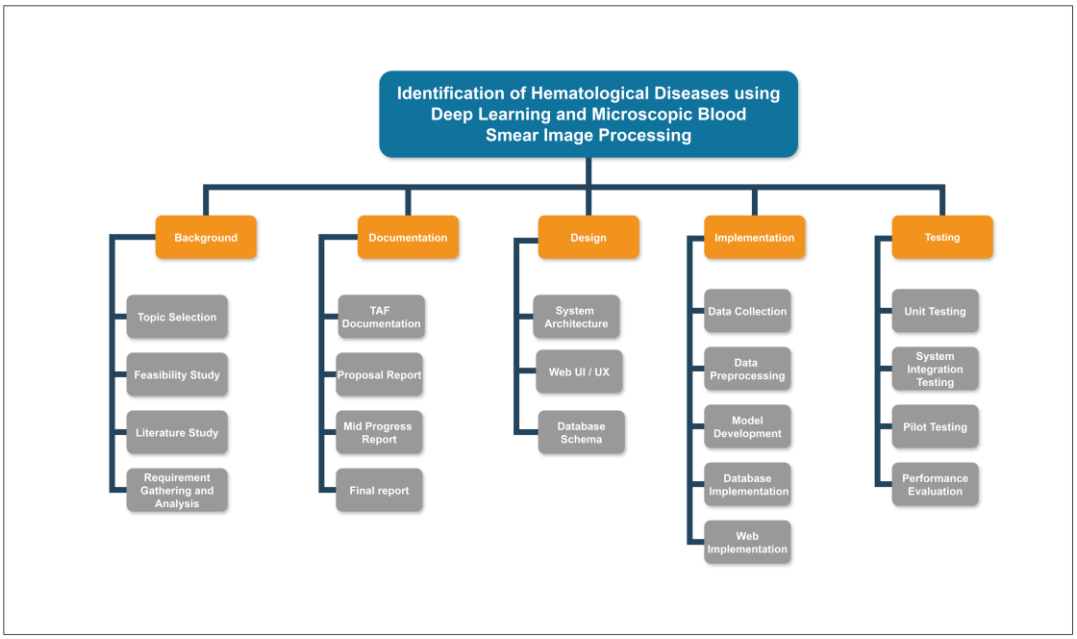
5. APPENDICES

5.1 Appendix A: Work Breakdown Chart



5.2 Appendix B: Gantt Chart

Hemora - Identification of Hematological Diseases using Deep Learning and Microscopic Blood Smear Image Processing

Project ID: 25-26J-514

Project Proposal Report

Sachinthanee M A S – IT22326140

BSc (Hons) Degree in Information Technology Specializing in Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology,
Sri Lanka

August 2025

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Sri Lanka

August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Sachinthanee M A S	IT22326140	Sayuri

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

---------- 29/08/2025 ----------

Signature of the Supervisor: Date:
(Dr. Dinuka Wijendra)

---------- 29/08/2025 ----------

Signature of the Supervisor: Date:
(Ms. Jenny Krishara)

ABSTRACT

The project Hemora – Identification of Hematological Diseases using Deep Learning and Microscopic Blood Smear Image Processing aims to improve the diagnosis of blood disorders in Sri Lanka through artificial intelligence. This component focuses on Iron Deficiency Anemia (IDA), which remains the leading cause of anemia in the country. Current methods, such as Complete Blood Count (CBC) reports and Peripheral Blood Smear (PBS) analysis, are slow, subjective, and heavily reliant on specialists, creating delays in patient care.

The proposed system develops a hybrid AI-based diagnostic tool that integrates CBC indices (Hb, MCV, MCH, RDW) with PBS image features (microcytosis, hypochromia, anisocytosis, poikilocytosis). Data will be collected locally, ethically approved, and validated by hematologists. A fusion of deep learning for image analysis and machine learning for CBC data will produce accurate and rapid IDA predictions. Deployed as a web-based platform, the system aims to provide reliable diagnostic support, reduce delays, and strengthen healthcare delivery in both urban and rural hospitals in Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Description
WHO	World Health Organization
IDA	Iron Deficiency Anemia
CBC	Complete Blood Count
PBS	Peripheral Blood Smear
CNN	Convolutional Neural Networks
SCD	Sickle Cell Disease
MCV	Mean Corpuscular Volume
MCH	Mean Corpuscular Hemoglobin
Hb	Hemoglobin
RDW	Red Cell Distribution Width
ANN	Artificial Neural Network

1. INTRODUCTION

The research project Hemora – Identification of Hematological Diseases using Deep Learning and Microscopic Blood Smear Image Processing is designed to provide a contemporary solution for the early and accurate detection of blood related disorders. Hematological diseases such as Iron Deficiency Anemia (IDA), Leukemia, Sickle Cell Disease (SCD) and Thalassemia are commonly detected using blood smear microscopy. But the procedure is tedious and highly dependent on the availability of skilled professionals.

This reliance on manual interpretation creates delays in patient care in many Sri Lankan hospitals, especially those in the northern and rural areas [1]. By integrating deep learning with microscopic images and CBC report processing, Hemora aims to provide an automated diagnostic system that improves the reliability, accessibility, and speed of diagnosing hematological diseases [6, 7, 8].

Among these conditions, IDA is unique because of its great prevalence and severe clinical and financial effects. IDA affects the physical and cognitive development of children, reduces academic achievement, causes persistent tiredness and decreases productivity in adults. Early births, maternal issues, and low birth weight kids are among the risks it raises for women, especially during pregnancy. The development of quick and precise diagnostic techniques is of the highest priority since these effects go beyond personal health to impact families, communities, and national production.

Peripheral blood smear (PBS) analysis and Complete Blood Count (CBC) data can be combined to diagnose IDA in healthcare environments. Low hemoglobin (Hb) decreased mean corpuscular volume (MCV), and decreased mean corpuscular hemoglobin (MCH) are indicative of microcytic, hypochromic anemia in CBC in patients with IDA. Although IDA cannot be completely separated from other microcytic anemias, these numerical values offer a quantitative indicator. In addition, PBS images show the typical morphological abnormalities of red blood cells, such as poikilocytosis (abnormally shaped

cells like pencil cells), anisocytosis (variation in cell size), hypochromia (pale central pallor due to reduced Hb content), and microcytosis (small cell size).

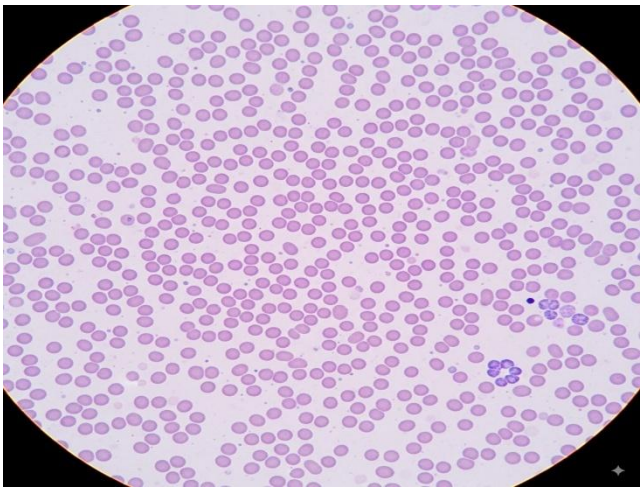

Figure 1.1: IDA blood cell

Hospitals in Sri Lanka currently use peripheral blood smear (PBS) analysis and complete blood count (CBC) tests separately to diagnose IDA. Although CBC offers quantitative indicators such red cell characteristics and hemoglobin levels, it is unable to objectively verify IDA. Ignoring the subjective and tedious nature of assessment, PBS enables medical professionals to visually recognize abnormal red blood cell morphologies such microcytosis, hypochromia, anisocytosis, and poikilocytosis. Furthermore, it can be difficult to make fast treatment decisions when there are few resources available, which frequently delay results.

This Hemora component suggests developing a hybrid AI-based diagnostic system that combines PBS image attributes and CBC indices in order to overcome these difficulties. In comparison with single-modality techniques, this system will integrate both qualitative and quantitative data, making it possible to separate IDA from other circumstances with more reliability. The system will provide healthcare professionals with an easily accessible tool to enable faster, more accurate, and more affordable IDA detection by testing the model using Sri Lankan datasets and implementing it as a web-based

application. This could improve the standard of care in impoverished rural hospitals as well as urban hospitals.

1.1 Background and Literature Survey

IDA is a recognized problem in Sri Lanka as several studies have shown that that it is the main cause of anemia in specific communities. Based on the *National Nutrition and Micronutrient Survey (2019)*, iron deficiency was the main cause of anemia in 15% of children under five and 36% of women of reproductive age [1]. The WHO estimates that approximately 29% of pregnant women in Sri Lanka suffer from anemia, showing that pregnant women are particularly impacted [2]. Importantly, research shows that 30%–50% of all cases of anemia in Sri Lanka are caused by IDA. This is equivalent to around 10%–20% of preschool children, 20%–25% of women who are of reproductive age, and around 6%–10% of pregnant women who have IDA specifically [1]. These figures indicate the urgent requirement for scalable, efficient diagnostic tools to reduce the local IDA burden.

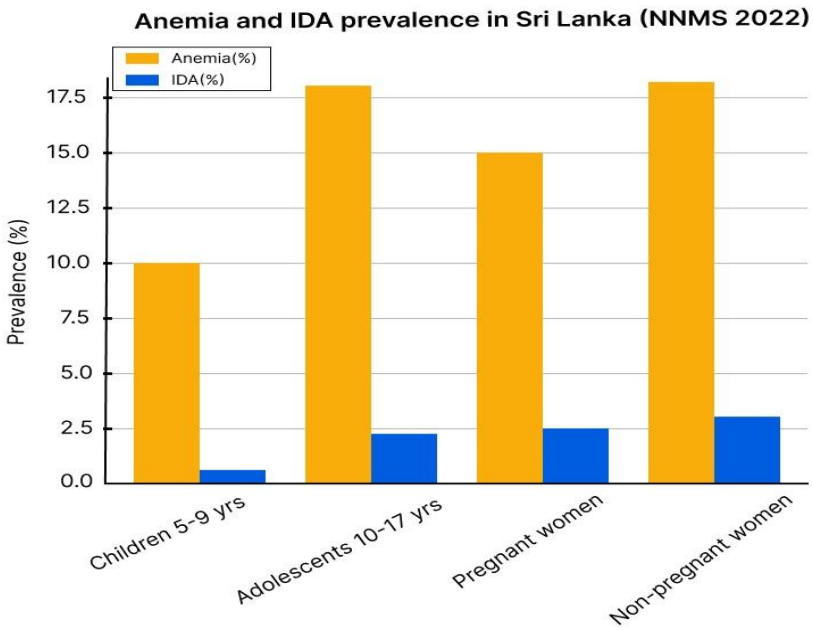

Figure 1.2:Anemia and IDA prevalence in SL

Complete Blood Count report testing and manual blood smear analysis are currently used in Sri Lankan diagnostic procedures. Quantitative indications of anemia are provided by CBC report values such as Hb, MCV, MCH. However, CBC values by themselves are unable to accurately differentiate IDA from other microcytic anemia, including thalassemia trait, which is also very common in Sri Lanka. Peripheral blood smear (PBS) examination is also considered to detect morphological abnormalities such as microcytosis, hypochromia, anisocytosis, and poikilocytosis [1]. The lack of qualified hematologists in many Sri Lankan organizations, particularly in rural areas, causes diagnostic delays of 3–7 days in government hospitals, which makes prompt intervention difficult [1].

Researchers in Sri Lanka have already established innovation in this field of research. Locally developed HemoSmart is a simple mobile technology for hemoglobin estimation based on machine learning [4]. Additionally, Karunasena et al. demonstrated the potential of smear-based AI solutions in Sri Lanka by applying deep learning (R-CNN) to peripheral smear images and obtaining over 90% accuracy in red blood cell classification [5]. Despite these developments, there is currently no locally verified hybrid system that combines PBS morphology and CBC data into a single system that allows for the fast and precise identification of IDA.

Globally, there are several AI driven approaches that have been developed to address IDA diagnostics by combining CBC statistics with smear image analysis. In order to detect poikilocyte cells linked to IDA, Tyagi et al. (2016) used an Artificial Neural Network (ANN) and were able to reach an accuracy of over 80% [6]. Hortinela et al. (2019) achieved 93% accuracy in classifying anemia related variables by applying Support Vector Machine (SVM) algorithms on RBC morphological features (area, perimeter, and geometric ratios) [7].

Convolutional Neural Networks (CNNs) have been proven by Macawile et al. (2018) to be capable of successfully identifying blood cell pictures with little feature engineering

[8], showing the great potential of deep learning techniques. The potential of AI in blood diagnostics is demonstrated by these international developments, which also point to areas where Sri Lanka could adapt.

A Sri Lankan system combining CBC data and Peripheral Blood Smear images could speed up IDA detection, ensure consistent smear analysis across organizations, and support national plans.

1.2 Research Gap

Sri Lanka lacks a hybrid diagnostic approach that combines CBC indices with peripheral blood smear (PBS) morphology for the identification of Iron Deficiency Anemia (IDA), despite strong national data and recent AI based studies. Especially in rural hospitals where smear analysis causes delays, current diagnostic procedures are still time-consuming and manual [1].

Local innovation, like HemoSmart [4], has demonstrated that Sri Lanka has the technical ability to develop safe machine learning based hemoglobin estimation methods. However, this technique is unable to accurately identify IDA since it does not evaluate the shape of red blood cells. Similarly, Karunasena et al. [5] used R-CNN to show AI driven RBC categorization on Sri Lankan blood smear images. However, this study was only proof of concept and did not incorporate CBC parameters into the diagnosis procedure.

Several AI methods have been used for blood diagnosis on an international level. Hortinela et al. [7] used SVM to categorize abnormal RBC morphologies, whereas Tyagi et al. [6] used ANN to identify poikilocytes in IDA. Both methods showed high accuracy. Furthermore, Macawile et al. [8] showed the overall strength of deep learning in blood picture analysis by using CNN to classify and count white blood cells. All of this research, however, focused on non-IDA situations or single modalities (blood smear image characteristics or CBC alone). None developed a single, locally verified diagnostic procedure that included PBS morphology and CBC indices.

This creates an obvious research gap for clinical practice to be accurate, quick, and economical, Sri Lanka needs a system designed for IDA that combines quantitative (CBC) and qualitative (PBS morphology) elements.

Table 1.1:Research gap table

Specification	Uses CBC?	Uses PBS Morphology?	IDA-specific?	Sri Lanka-specific?	Validated/Robust?
White Blood Cell Classification using CNN [8]	No	No (focuses on WBCs)	No	No	Yes (but not for IDA)
HemoSmart: ML Device & Mobile App [4]	Yes (Hb estimation only)	No	No	Yes	Yes (but limited to Hb)
Detection of Poikilocyte Cells in IDA using ANN [6]	No	Yes	Yes	No	Partially (accuracy >80%)
Identification of Abnormal RBCs using SVM [7]	No	Yes	No (general RBC shapes)	No	Yes (high accuracy)
Proposed Component (Sri Lanka)	Yes	Yes	Yes	Yes	To be validated

1.3 Research Problem

According to national surveys and studies, 30–50% of all anemia cases in children, women of reproductive age, and pregnant women are caused by iron deficiency anemia (IDA), which continues to be a persistent public health concern in Sri Lanka [1], [2], and [3]. Due to the significant reliance on manual interpretation and a lack of hematologists, especially in rural hospitals, diagnosis is frequently delayed and inconsistent even with the

availability of peripheral blood smear (PBS) microscopy and Complete Blood Count (CBC) testing [1].

Though they are still only one-mode systems, recent innovations from Sri Lanka show potential. A lightweight, non-invasive machine learning device called HemoSmart [4] can estimate hemoglobin levels but cannot record PBS morphology. Similar to this, Karunasena et al. [5] used R-CNN on PBS images and classified red blood cells with more than 90% accuracy .However, the method was not verified as a complete IDA diagnostic tool or combined with CBC indices.

The potential of AI in hematology has been confirmed by global studies. Hortinela et al. [7] implemented SVM to categorize abnormal red cells with excellent accuracy, Tyagi et al. [6] used ANN for poikilocyte cell detection in IDA smears, and Macawile et al. [8] used CNN to count and classify white blood cells. Although these studies show the technical feasibility of AI-driven diagnoses, they have been conducted out on non-local datasets outside of Sri Lanka and cannot be immediately implemented in hospitals there.

As of right now, no hybrid diagnostic approach that combines PBS morphological features and CBC indices for IDA identification has been validated in Sri Lanka. Current solutions are either image based but insufficient (R-CNN), non-invasive but limited (HemoSmart), or foreign models that haven't been adjusted for local datasets and workflows.

2. RESEARCH OBJECTIVES

2.1 Main Objective

To develop and establish a diagnostic system that combines PBS image features with CBC data in order to accurately, quickly, and affordably diagnose IDA in medical environments in Sri Lanka.

2.2 Specific Objective

1. To examine, combining data from national surveys and local research, the prevalence, diagnostic techniques, and limitations of Sri Lanka's existing IDA detection procedures.

2. To develop a hybrid diagnostic approach combining qualitative morphological features from PBS pictures with quantitative CBC parameters.

3. To apply deep learning models to identify red blood cell abnormalities specific to IDA, such as hypochromia, microcytosis, anisocytosis, and poikilocytosis.

4. To validate model by using Sri Lankan patient datasets to compare its accuracy, sensitivity, and specificity to actual expert diagnoses

5. To determine whether the system can be integrated into Sri Lankan hospital workflows as a web-based diagnostic support tool that helps medical professionals in rural areas with limited resources and reduces down on diagnostic delays.

3. METHODOLOGY

3.1 overview

The research's method focuses on creating a diagnostic system that has been validated in Sri Lanka for the identification of iron deficiency anemia (IDA) by combining PBS images with CBC data. Digital CBC reports and stained microscopic smear images will be gathered from nearby hospitals and labs, and hematologists will annotate the images to offer ground truth. Quantitative features will be provided by CBC indices like hemoglobin Hb, MCV, MCH, and red cell distribution width (RDW), while qualitative abnormalities such as microcytosis, hypochromia, anisocytosis, and poikilocytosis will be captured by PBS images.

Both CBC data and PBS images will undergo preprocessing prior to analysis. While images will be subjected to color normalization, noise reduction, red blood cell segmentation, and image patch extraction to enhance quality and consistency, CBC values will be normalized to remove scale inconsistencies. Following that, features will be extracted from both modalities. CBC indices will directly represent blood parameters, while image features will be derived by combining deep learning-based representations using Convolutional Neural Networks (CNNs) with manually created descriptors (like size, shape, and color intensity).

The architecture will be based on a hybrid concept. While a model will be developed on PBS pictures to characterize red blood cell morphologies, a machine learning classifier will initially be taught on CBC indices to identify anemia trends. Either feature level fusion, which concatenates CBC and picture features into a single vector, or decision level fusion, which weights and combines distinct predictions, will then be used to merge these two outputs. The model may use both morphological and quantitative information for precise IDA detection because to this integrated methodology.

Annotated datasets will be used for model training, while independent validation sets and cross validation will be used for hyperparameter optimization and validation. The evaluation measures of accuracy, sensitivity, specificity, F1-score, and AUC-ROC will be used to test the final performance on unknown data. The model will be made available online as a diagnostic tool after it has been validated. Through a straightforward interface, clinicians will be able to input PBS images and CBC data. The system will then provide a structured diagnostic report that includes annotated smear images, aberrant red blood cell percentages, critical CBC indices, and the anticipated outcome.

Usability will be given top priority throughout deployment in Sri Lankan hospital workflows, particularly in government facilities with a shortage of specialists. Prior to data collection, ethical approval will be obtained, and all patient information will be protected and anonymized. Faster, more accurate, and more economical IDA identification in both urban and rural healthcare settings will be made possible by the tool, which will act as a clinical decision-support system, supporting but not replacing medical experts.

3.2 System Architecture

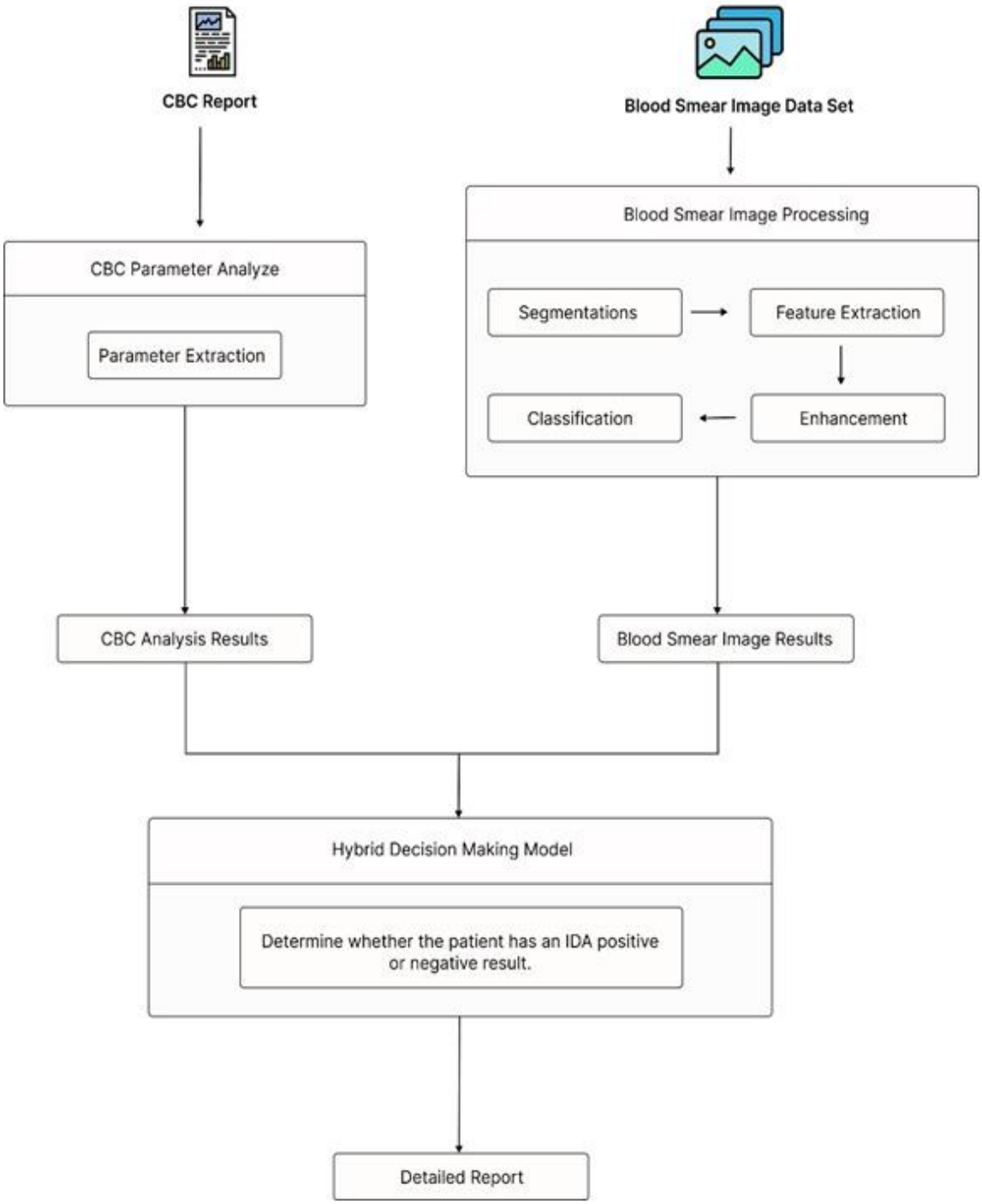

Figure 3.1:System Architecture

3.3 Tasks and Sub Tasks

Task 1: Data Collection

Sub-task 1.1: Obtain peripheral blood smear images from Sri Lankan hospitals and diagnostic centers

Sub-task 1.2: Gather Complete Blood Count (CBC) reports from the same patients

Sub-task 1.3: Secure ethical clearance and ensure patient anonymity

Task 2: Data Preprocessing

Sub-task 2.1: Apply noise reduction and contrast enhancement to images

Sub-task 2.2: Perform cell segmentation using U-Net or watershed algorithms

Sub-task 2.3: Implement data augmentation (rotation, flipping, scaling)

Sub-task 2.4: Normalize CBC values and organize into feature vectors

Sub-task 2.5: Link patient blood smear features with CBC parameters

Task 3: Model Development

Sub-task 3.1: Develop a deep learning architecture for lymphoblast morphology recognition

Sub-task 3.2: Create machine learning classifiers for CBC parameter analysis

Sub-task 3.3: Design hybrid fusion system combining image and CBC models

Task 4: Validation

Sub-task 4.1: Validate system performance using real-world clinical data

Sub-task 4.2: Calculate performance metrics (accuracy, sensitivity, specificity, precision)

Sub-task 4.3: Perform cross validation to ensure model generalizability

Sub-task 4.4: Compare predictions against bone marrow biopsy ground truth data

Task 5: Deployment

Sub-task 5.1: Develop web-based platform architecture

Sub-task 5.2: Create upload interface for blood smear images

Sub-task 5.3: Design input fields for CBC parameters

Sub-task 5.4: Implement automated diagnostic suggestions and downloadable reports

Task 6: Evaluation in Clinical Settings

Sub-task 6.1: Conduct pilot testing with Sri Lankan healthcare providers

Sub-task 6.2: Collect feedback from laboratory personnel and physicians

Sub-task 6.3: Plan integration with hospital information systems (HIS)

3.4 Materials and Resources

Hardware Resources

- Digital microscopes or slide scanners for capturing PBS images.
- Secure data storage systems for CBC records and image datasets.

Software Resources

- Programming languages: Python.

Human Recourses

- Hematologists for data annotation and diagnostic validation.
- Software engineers for model development and deployment.
- Clinicians and laboratory technicians for pilot testing and usability feedback.
- System administrators for maintaining the deployed platform in hospitals.

3.5 Data requirements and collection

Data Needed

- CBC Parameters: Hb, MCV, MCH, RDW
- PBS Images: Stained blood smear images showing RBC morphologies such as microcytosis, hypochromia, anisocytosis, and poikilocytosis.

Data Sources

- Sri Lankan hospitals and diagnostic laboratories

Collection Method

- CBC data extracted from patient reports.
- PBS images are captured using digital microscopes or slide scanners.
- Annotation of PBS images by qualified hematologists to provide labels.
- CBC data and PBS images are linked securely for each patient to form a hybrid dataset

Ethical Considerations

- Ethical clearance obtained from relevant review committees.
- Patient consent secured where applicable.
- Data anonymized and encrypted to maintain confidentiality.

Dataset Preparation

- Data split into training, validation, and testing sets.

3.6 Feasibility Study

The proposed system's potential, effectiveness, and implementation in Sri Lanka's healthcare system would be determined by the feasibility study.

- **Technical Feasibility**

Since AI and machine learning algorithms like CNNs, ANNs, and SVMs have already been successfully used in blood cell classification and anemia detection, the proposed approach is possible in theory. These techniques can be modified for a hybrid framework if CBC data and enhanced PBS images are available. Current local breakthroughs like R-CNN smear analysis [5] and HemoSmart [4] attest to the availability of the necessary knowledge and technology in Sri Lanka, making use technically feasible.

- **Operational Feasibility**

From an operational point of view, the system can easily fit into the way hospitals already work without causing major changes. Doctors and lab technicians will simply need to upload CBC results and blood smear images through a straightforward web interface. Within a few minutes, the system will generate a clear diagnostic report, saving valuable time. This helps overcome the long delays currently faced in blood smear interpretation due to the shortage of hematologists, especially in rural hospitals, and makes patient care faster and more efficient.

- **Economic Feasibility**

The system's future cost savings provide evidence for its financial sustainability. The automation of diagnostic procedures will decrease the reliance on expert interpretation and save time and money, even though the initial expenditure includes system development, implementation, and user training. Indirectly saving healthcare expenses for

patients and institutions, quicker identification of IDA can also shorten hospital stays and treatment delays.

- **Legal and Ethical Feasibility**

The system will strictly follow medical data protection standards to safeguard patient privacy. All patient information will be anonymized, encrypted, and stored securely to maintain confidentiality. Ethical approval will be obtained before collecting any data, and the system will operate only as a clinical decision-support tool, ensuring that the final responsibility for diagnosis always rests with medical professionals. In this way, the technology is designed to support and complement human expertise, not replacing it, while fully complying with legal and ethical requirements.

4. PROJECT REQUIRMENTS

4.1 Functional Requirements

- User Authentication - The system must provide a secure login for clinicians and laboratory staff to ensure that only authorised users can access patient data. It should also support role-based access so that each user can only perform tasks relevant to their responsibilities.

- Data Input - The system must support the uploading of CBC report values such as Hb, MCV, MCH, and RDW along with digitised PBS images. This ensures that both numerical and morphological data are available for hybrid analysis of IDA.

- Data Preprocessing - The system must normalise CBC values to ensure consistency across different laboratory machines and reporting formats. It should also preprocess PBS images through segmentation, enhancement, and resizing to improve the quality of data used for analysis.

- Feature Extraction - The system must extract features from CBC (numerical indices) and PBS images (cell morphology, microcytosis, hypochromia, anisocytosis, poikilocytosis).

- Hybrid Model - The system must fuse CBC indices and PBS image features using suitable machine learning or deep learning algorithms to achieve more reliable predictions. This hybrid approach will improve diagnostic accuracy by combining both quantitative and morphological evidence of IDA.

- Disease Classification - The system must identify whether a patient is affected by IDA or not by analysing the combined CBC and PBS features. It should also

provide probability scores or confidence levels to help clinicians understand the reliability of the prediction.

4.2 Non-functional Requirements

- Accuracy - The hybrid model must achieve a minimum of 90% sensitivity to ensure that true IDA cases are correctly identified. It should also maintain at least 85% specificity to reduce false positives and avoid misdiagnosing patients without IDA.

- Scalability - The system should be able to accommodate multiple users simultaneously, allowing clinicians and lab staff from different healthcare departments to access it without delays. This ensures smooth operation in busy hospital environments where several cases may be processed at the same time.

- Usability - The system should provide a simple and intuitive interface so that doctors with minimal technological expertise can easily navigate it. Clear instructions, tooltips, and a user-friendly design will ensure that clinical staff can use the system without extensive training.

- Security - To comply with healthcare data privacy regulations, all patient data must be secured during both transmission and storage. This can be achieved through strong encryption methods and access control, ensuring that sensitive medical information remains confidential and protected from unauthorized use.

- Maintainability - The system should allow straightforward updates to the AI model whenever new data becomes available. This ensures that the model can be retrained easily, improving accuracy over time and keeping the diagnostic system aligned with current clinical needs.

- Performance - The system must be capable of processing each case and generating a diagnostic report in under two minutes. This quick turnaround will help clinicians make timely decisions, reducing delays in patient care and treatment.

4.3 User Requirements

- Ease of use - Users require a straightforward interface that allows them to upload CBC values and PBS images without dealing with technical complexity. This ensures that clinicians and lab staff can focus on patient care rather than navigating complicated software.

- Rapid results - Users expect the system to provide diagnostic results within a short time frame so that it can effectively support clinical decision-making. Fast output is especially important in hospital environments where timely treatment can significantly improve patient outcomes.

- Accuracy and reliability - Users require the system to generate results that are consistent with the interpretations of expert haematologists. This reliability builds trust in the tool and ensures that clinicians can confidently use it to support patient diagnosis and treatment.

- Visual Outputs - Clinicians expect the system to provide annotated PBS images that highlight detected abnormal cells for greater transparency. These visual outputs help medical staff verify the system's predictions and build confidence in its diagnostic support.

- Data security - Users require a straightforward interface that allows them to upload CBC values and PBS images without dealing with technical complexity. This ensures that clinicians and lab staff can focus on patient care rather than navigating complicated software.

- Error handling - Users need the system to provide clear fault notifications, such as warnings when a smear image is of poor quality or unreadable. Along with these alerts, the system should give simple guidance on how to correct the issue so that accurate results can still be obtained.

- Accessibility - The system should be designed so that it can be used effectively in both urban hospitals and rural healthcare facilities with limited resources. This ensures equal access to reliable diagnostic support across the country, regardless of location or infrastructure.

4.4 System Requirements

4.4.1 Hardware Requirements

- Microscope with camera attachment
- Computer with GPU (e.g., NVIDIA RTX 2060 or higher)
- Minimum 16GB RAM
- Intel i7 or equivalent processor

4.4.2 Software Requirements

- Operating System: Windows 10 or Ubuntu 20.04
- Programming Language: Python 3.9 or above
- Deep Learning Frameworks: TensorFlow / PyTorch.
- Image Processing Libraries: OpenCV, Scikit-image.
- Database: MongoDB / PostgreSQL
- Web Framework: Flask / Django (for backend), React / Angular (for frontend)

5. DESCRIPTION OF PERSONAL AND FACILITIES

5.1 Personals

The project is being conducted by a group of four final-year undergraduate students who specialize in software engineering. Each member builds a unique component of the overall system, aligned with their experience.

- Member 1 - Acute Lymphoblastic Leukemia (ALL) Component: Responsible for developing the component for the detection and identification of Acute Lymphoblastic Leukemia (ALL).
- Member 2 - Beta-Thalassemia (Minor) Component: Responsible for developing the component for the detection and identification of Beta-Thalassemia (Minor)
- Member 3 - Sickle Cell Disease (SCD) Component: Responsible for developing the component for the detection and identification of Sickle Cell Disease (SCD)
- Member 4 - Iron Deficiency Anemia (IDA) Component: Responsible for developing the component for the detection and identification of Iron Deficiency Anemia (IDA)

The team collaborates on data preprocessing, model integration, testing, and web-based system deployment to ensure that all components are integrated into a single, unified platform.

5.2 Facilities

The research will utilize the following facilities and resources:

- University laboratories are fitted with high-performance computers and internet access to facilitate model training, testing, and development.

- Software Resources: TensorFlow, PyTorch, and Scikit-learn are open-source frameworks for deep learning and machine learning model construction.

- Web development tools include React.js, Node.js, and database management systems for creating the web-based platform.

- Data Sources: We worked with hospitals, laboratories, and publicly available datasets to acquire anonymized blood smear images and CBC results.

- Cloud computing resources include the use of platforms such as Google Colab Pro or AWS for GPU-accelerated model training when required.

- Ethical and organizational support: University supervisors provide guidance on gaining approvals and ethical clearance for clinical data collecting in Sri Lanka.

The combination of qualified individuals and accessible facilities ensures that the project is successfully finished within the academic schedule.

6. BUDGET AND BUDGET JUSTIFICATION

Table 6.1:Budget Table

Item	Description	Estimated Cost (LKR)
Cloud GPU/Computing	Occasional use of Colab Pro+ or similar platforms for model training	15,000
Web Deployment	Domain name registration and hosting for the system	8,000
Data Collection & Logistics	Printing, documentation, and travel for hospital visits	7,000
Miscellaneous	Stationery, internet, contingency expenses	5,000
Total		**35,000**

6.1 Budget justification

- Cloud GPU/Computing: Free services have limited GPU support. therefore a minimal allocation is created for efficient training during peak experiments.
- Web Deployment: A domain name and basic hosting are required to test the system in a clinical setting.
- Data Collection and Logistics: This includes minimal transportation expenditures to hospitals/clinics, as well as the printing of consent forms and ethical clearance documentation.
- Miscellaneous: Includes modest expenses like stationery and internet data.

APPENDICES

Appendix A: GANTT CHART

Figure A.1:Grantt Chart

Appendix B: Work Breakdown Structure

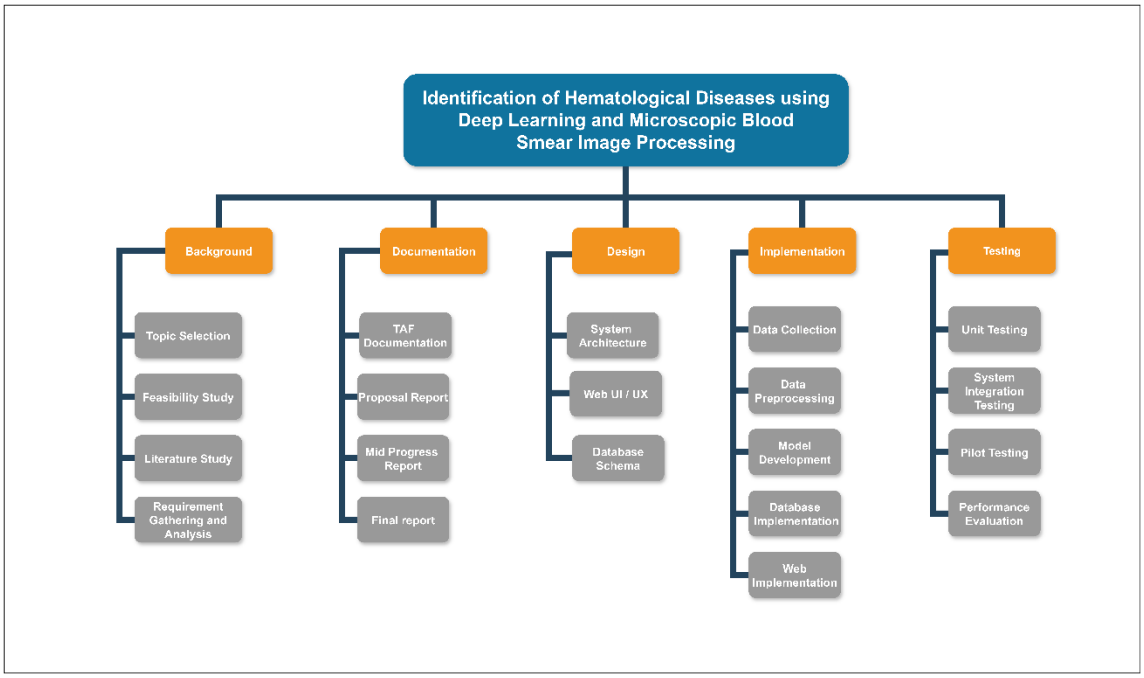

Figure B.1 : Work BreakDown Structure

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